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Perspective

The Mycobactin Biosynthesis Pathway: A Prospective Therapeutic Target in the Battle against Tuberculosis

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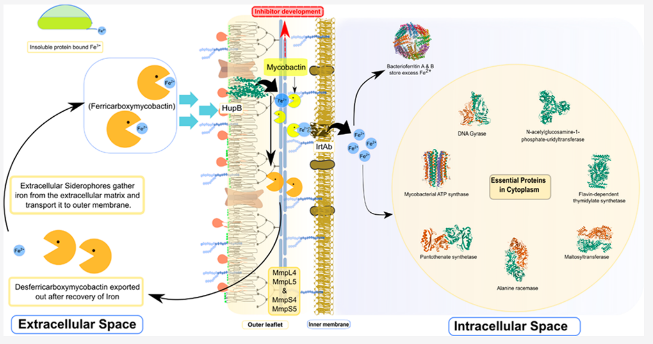

ABSTRACT: The alarming rise in drug-resistant clinical cases of tuberculosis (TB) has necessitated the rapid development of newer chemotherapeutic agents with novel mechanisms of action. The mycobactin biosynthesis pathway, conserved only among the mycolata family of actinobacteria, a group of intracellularly surviving bacterial pathogens that includes *Mycobacterium tuberculosis*, generates a salicyl-capped peptide mycobactin under iron-stress conditions in host macrophages to support the iron demands of the pathogen. This *in vivo* essentiality makes this less explored mycobactin biosynthesis pathway a promising endogenous target for novel lead-compounds discovery. In this Perspective, we have provided an up-to-date account of drug discovery efforts targeting selected enzymes (MbtI, MbtA, MbtM, and PPTase) from the *mbt* gene cluster (*mbtA-mbtN*). Furthermore, a succinct discussion on non-specific mycobactin biosynthesis inhibitors and the Trojan horse approach adopted to impair iron metabolism in mycobacteria has also been included in this Perspective.

1. INTRODUCTION

1.1. Clinically Approved Anti-tubercular Chemotherapeutics and Their Evolving Drug Resistance Mechanism. Tuberculosis (TB) is an ancient infectious disease caused by the deadly pathogen *Mycobacterium tuberculosis* (Mtb).[1,2] This bacterium has plagued humanity for centuries and is now the cause of a global pandemic.[3,4] According to the 2020 World Health Organization (WHO) report, TB was responsible for about 1.4 million deaths, and over 10 million fell ill with TB worldwide in 2019, implying its dreadful impact on the global mortality and morbidity rate.[5] The WHO estimates that about one in four individuals has an established TB infection as of date.[6] Other *Mycobacterium* genus members, such as *Mycobacterium bovis* and *Mycobacterium africanum*, are also cumulatively affecting the world population.[7] TB is an airborne disease and, as most of the infections of this kind, does not reach the active phase immediately after reaching the target. Due to its ability to adapt to the human immune system, Mtb survives latently infected individuals in a non-replicating or dormancy-like state for an indefinite period, reactivating only when there is an immune imbalance.[8] Bacillus resistance during the latent phase is accomplished by producing reactive nitrogen intermediates (RNIs), e.g., NO, NO_2, and N_2O_3, blocking phagosome−lysosome fusion, and interfering with MHC class II antigens.[9] Activation of the resuscitation promotion factors,[10,11] i.e., Rpf A to E (peptidoglycan-hydrolyzing enzymes), enables Mtb to re-enter the active phase, whereas the Mtb DosR regulon system, *dosR* (Rv3133c),[12,13] enables a transition from respiring to non-respiring conditions and vice versa without loss of viability, jointly leading to active and contagious TB infection.[14] These salient features of the bacilli make TB one of the most challenging infectious diseases to combat and eradicate.[15,16] The golden era of TB antibiotic discovery commenced with the *p*-aminosalicylic acid development in 1943, followed by the discovery of streptomycin as the first anti-tubercular agent in 1944.[17−19] This was followed by a series of anti-tubercular drug discoveries, including isoniazid and pyrazinamide in 1952, ethambutol in 1961, and rifampicin in 1963, with the latter remaining as the keystone for anti-TB therapy to date.[20−24] (Figure 1)

In the mid-1980s, a significant resurgence of TB infections occurred primarily due to the prevalence of extensively drug-resistant (XDR) or multi-drug-resistant (MDR) strains, which were potentially incurable with available therapies.[25] In 1993,

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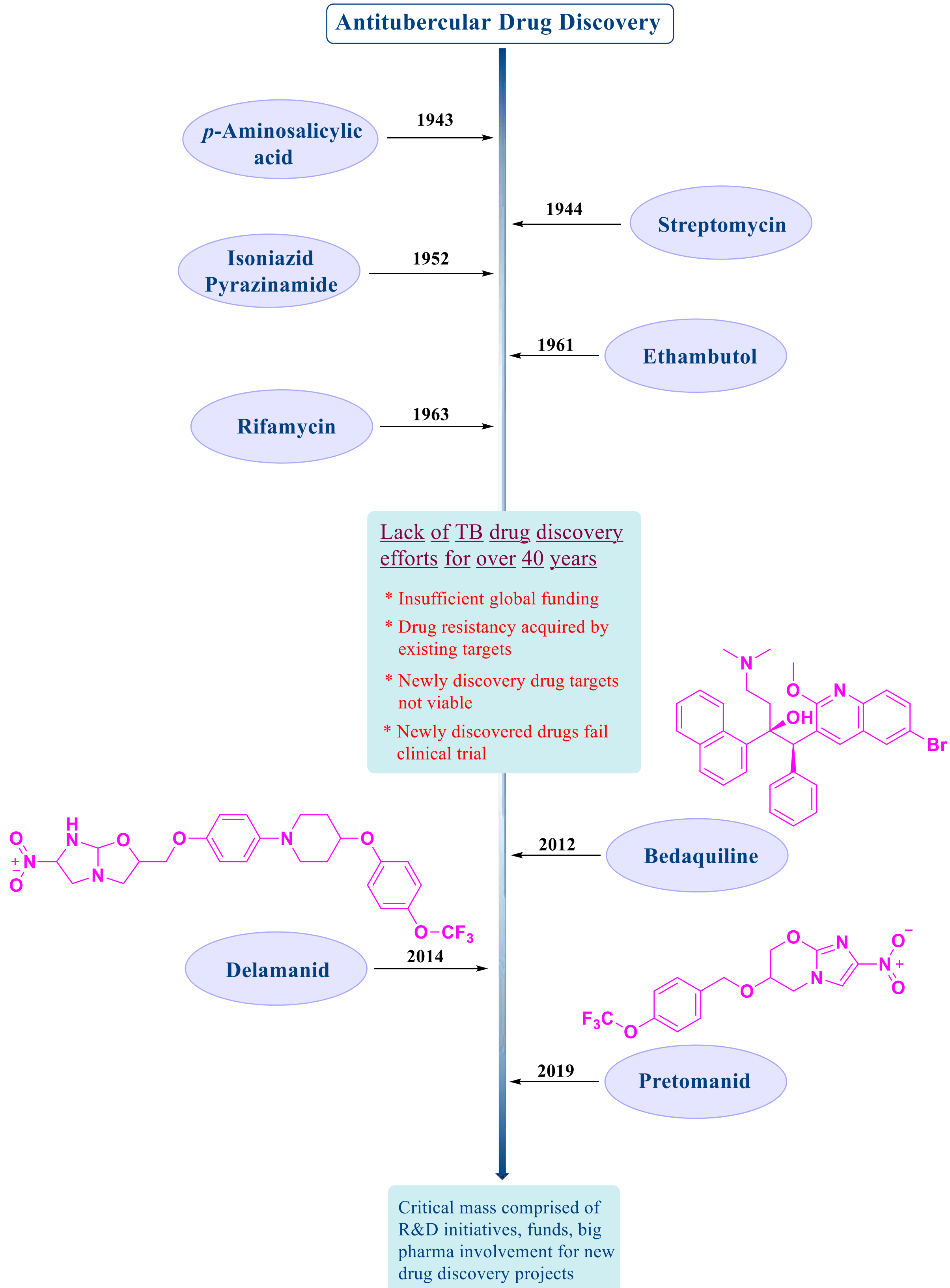

Figure 1. Schematic description of the chronological discovery sequence of clinically approved anti-tubercular drug candidates, starting with the *p*-aminosalicylic acid and moving to the recently approved pretomanid, indicating the progress in TB management over 76 years.

the WHO declared TB a global health emergency to draw the world's attention to the growing severity of the TB epidemic.[26] Due to the continuous evolution of essential genetic material and the rapid propagation of drug-resistant TB pathogens, the therapeutic efficacy of the drugs reduced drastically over time. Moreover, the lack of investment in R&D from big

Table 1. Summary of Drugs, Targets, and Common Resistance Mechanisms for Clinically Approved Anti-tubercular Agents

drug	discovery/ approval	target	resistance mechanism
p-aminosalicylic acid (PAS)	1943	The mechanism of action of PAS remains elusive, although it is thought that this drug might act as an anti-metabolite of salicylic acid and be involved in some steps of the metabolic pathway of iron uptake.[38] More recently, PAS mode of action has been linked to the disruption of folate metabolism in Mtb, as this prodrug was found to target dihydrofolate reductase (DHFR) and modulate thymidylate synthase (*thyA*) activity.[39]	Mutation in the *thyA* gene in addition to *folC* and *ribD* genes has been identified as the leading reason behind PAS resistance.[40,41]
streptomycin	1944	Inhibits protein synthesis by targeting S12 protein and 16S rRNA components of the 30S ribosomal subunit.[42]	Mutation at 30S ribosomal protein S12 (rpsL) and 16S rRNA (rrs) genes conferring binding site modulation.[43]
isoniazid	1952	Inhibits cell wall mycolic acid biosynthesis by targeting NADH-dependent enoyl acyl carrier protein reductase.[44]	Suppression of catalase-peroxidase (KatG) enzymes responsible for prodrug activation of INH result in a decrease of the active drug release, and mutation of inhibin Subunit Alpha (inhA) gene causes overexpression of inhA protein.[45]
pyrazinamide	1952	Inhibits membrane energetics and phthiocerol dimycocerosates (PDIM) synthesis by interfering with aspartate decarboxylase (PanD), mycocerosic acid synthase, and phenolthiocerol synthesis type-I polyketide synthases.[46]	Mutations in pyrazinamidase/nicotinamidase (pncA) gene reduce the conversion of pyrazinamide to active pyrazinoic acid form.[47]
ethambutol	1961	Inhibits cell wall arabinogalactan synthesis by targeting arabinosyl transferase enzyme.[48]	Mutation at arabinosyl transferase (embB306) codon of embB gene responsible for the synthesis of a critical structural component of the mycobacterial cell wall, arabinogalactan, and lipoarabinomannan.[49]
rifampicin	1963	Inhibits RNA synthesis by the β subunit of RNA polymerase.[50]	Mutation at the subunit of RNA polymerase (rpoB) gene causes conformational changes.[51]
bedaquiline	2012	Inhibition of mitochondrial ATP synthase.[52,53]	Mutation at the *atpE* gene and Rv0678 conferring drug resistance. *atpE* gene encodes the trans membrane oligomeric C subunit of ATP synthase, preventing the drug from binding at the C subunit, whereas Rv0678 is a transcriptional repressor encoding the MmpS5-MmmpL5 efflux pump involves in resistance to bedaquiline.[54]
delamanid	2014	Inhibits methoxymycolic acid and ketomycolic acid biosynthesis.[55]	Mutation of the five genes *ddn*, *fgd1*, *fbiA*, *fbiB*, and *fbiC*, associated with either prodrug activation or F_{420} biosynthetic pathway, has been identified as the reason for drug resistance.[56,57]
pretomanid	2019	Targeting the cell wall biosynthesis inhibition and causing respiratory poisoning through nitric oxide (NO) release.[58] Additionally, targeting the pentose phosphate pathway results in the accumulation of lethal methylglyoxal.[59]	Function mutation in *ddn*, *fgd1*, *fbiA*, *fbiB*, and *fbiC* causes drug resistance.[56,57]

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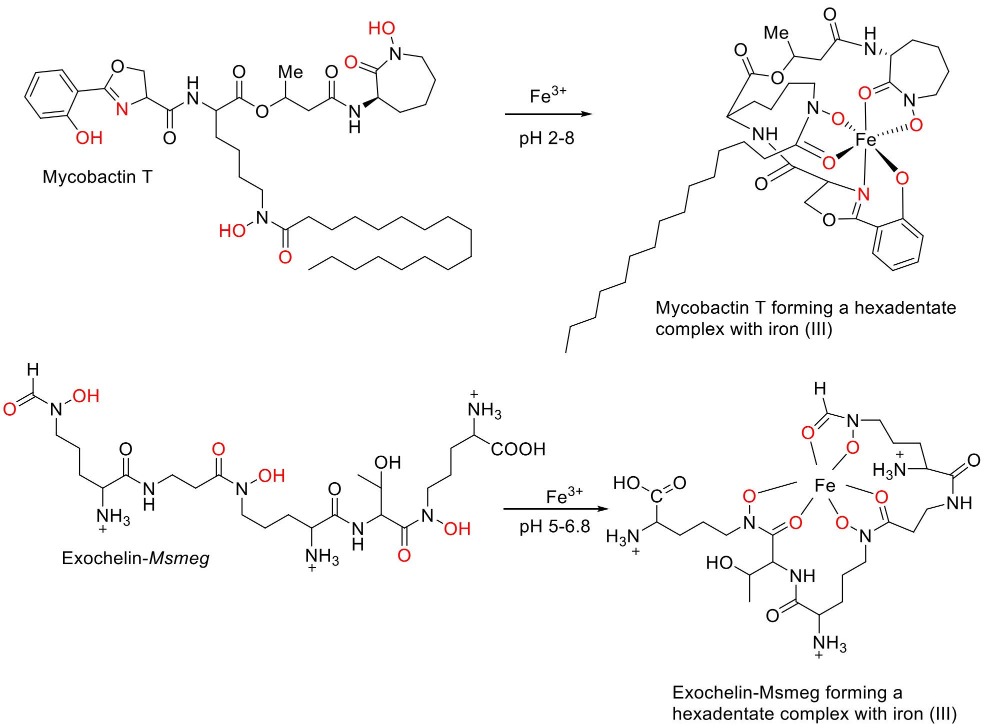

Figure 2. Mycobactin T and exochelin-*Msmeg* forming a hexadentate complex with iron(III) within a pH range of 2−8 and 5−6.8, respectively.

pharmaceutical companies in newer candidates' discoveries worsened over the past few decades. Clinical approval of new drugs, i.e., bedaquiline (2012),[27−29] delamanid (2014),[30,31] and pretomanid (2019),[32,33] came more than four decades after the discovery of rifampicin (more information is summarized at https://www.newtbdrugs.org/pipeline/discovery) (Figure 1). Like first-line TB drugs, the newly discovered drugs also targeted the products of essential genes (essential for the sustenance of the microorganism in any environment).[34] Although bedaquiline, delamanid, and pretomanid remain the only solution against drug-resistant TB to date, the slow emergence of drug-resistant cases to these new drugs forecasts a grim scenario in anti-tubercular chemotherapy. This also highlights the need for newer chemical entities with novel modes of action to mitigate the emerging burden of MDR, XDR, and totally drug-resistant (TDR) TB.[35−37]

We have summarized the established essential drug targets and the mechanism underlying the development of drug resistance for each target in Table 1. We believe that this will encourage medicinal chemists to search for alternative drug targets and investigate newer ways of tackling drug resistance.

1.2. Conditionally Essential Target, Mycobactin, and Its Role As a Significant Drug Platform for Drug-Resistant TB. There is an urgent need for new anti-tubercular agents, which act via novel mechanisms different from the presently available drugs. This can be achieved using the concept of "conditionally essential target" (CET)-based drug design. A good example of a conditionally essential process in *mycobacteria* would be the acquisition of iron, an essential nutrient for bacterial biochemical machinery,[60−62] from the human host reservoirs. The concentration of iron in our serum or body fluids is significantly low (approximately 10^{-24} M) to support bacterial colonization and growth.[63] The extracellular iron in the host body is primarily bound to various iron transport proteins, such as transferrin and lactoferrins, and the intracellular iron is sequestered by the "heme" group stored within the iron−sulfur cluster of the storage protein ferritin.[64,65] Mycobacteria have evolved several mechanisms to acquire iron when faced by scarcity within the host. The most prominent is the synthesis, secretion, and re-uptake of small molecules termed mycobactins (mycobacterial siderophores or iron chelators).[66−68] Mycobacteria acquire iron from their host cell through the mycobactin-mediated iron transportation system. This siderophore biochemical machinery is up-regulated when Mtb face iron-deficient conditions, as experienced by Mtb in macrophages, and is one of the major pathogenic determinants of TB.[69,70]

Mycobactins are hexadentate ligands with a tripodal architecture. Their basic scaffold consists of an *o*-hydroxyphenyloxazoline system attached to an acylated hydroxylysine residue esterified with a 3-hydroxybutyric acid unit. This linker is, in turn, tethered to a terminal cyclized hydroxylysine moiety forming a seven-membered lactam ring. Mycobactins exhibit a very high affinity toward the ferric ion Fe(III) as they form a stable binary complex, with a stability constant (K_s) of $\sim 10^{-25}$ (Figure 2).[71−75] Mtb produces two main types of siderophores, e.g., carboxymycobactin and mycobactin T. The hydrophilic carboxymycobactin (a highly soluble peptide molecule)[76−78] is normally released in the extracellular space. It chelates and internalizes iron as a ferri-carboxymycobactin complex at physiological pH (2−8) and transfers Fe(III) to the more lipophilic mycobactin-T[79] that is associated with the mycobacterial cell wall. Another class of siderophores, termed exochelins, are produced by saprophytic mycobacteria, such as *Mycobacterium smegmatis* (*Msmeg*)[80] or *Mycobacterium neoaurum* (*Mneo*).[81] Unlike carboxymycobactin and mycobactin (virulence factors), these are considered avirulent factors that indicate ongoing TB infections. Exochelin is a peptide-based, highly water-soluble molecule, and its hydrophilic nature facilitates sequestration of Fe(III) from the aqueous aerobic environment of the host tissues followed by

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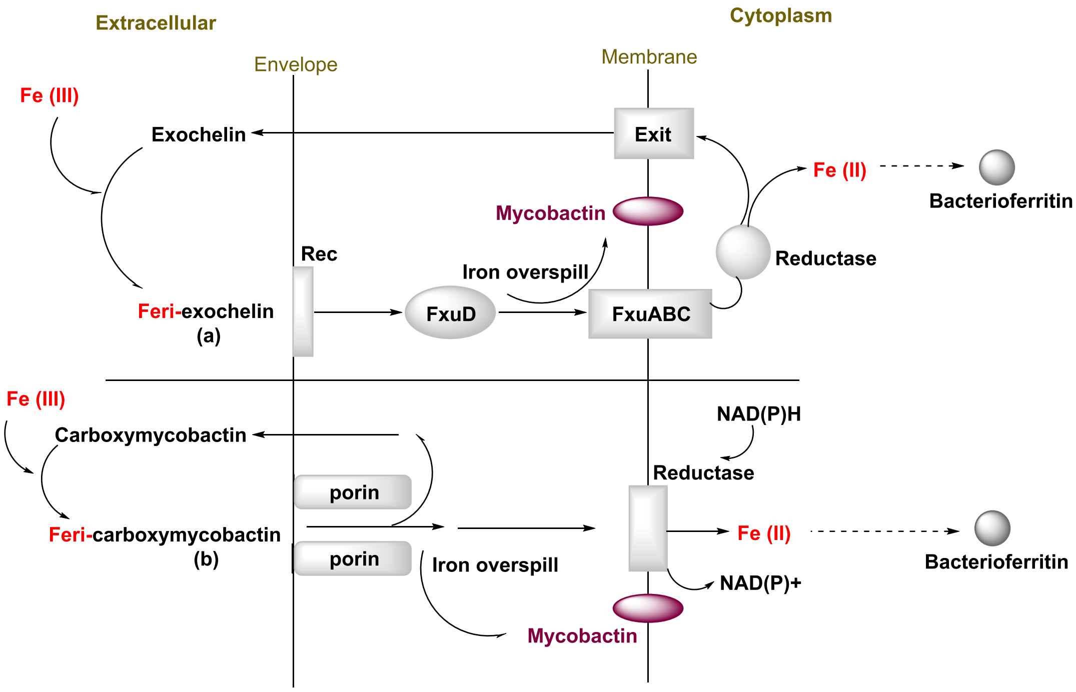

Figure 3. General diagrammatic representation of the different possible modes of iron uptake mechanism involving exochelin, mycobactin, and carboxymycobactin in mycobacteria. (a) Non-membrane-associated exochelins form a stable ferri-exochelin complex recognized by the surface receptor protein (Rec) that allows the complex to be taken across the cell envelope. Membrane penetration of the ferri-exochelin complex occurs through the FxuA, FxuB, and FxuC systems. Reductase enzymes present in the cytoplasm release Fe(II) ions, which are then incorporated into mycobacterial apoproteins, such as bacterioferritin (Bfr). (b) The ferri-carboxymycobactin uptake is facilitated by a diffusion mechanism through the membrane transport protein, porin. Membrane-associated reductase releases Fe(II) within the cytoplasm.

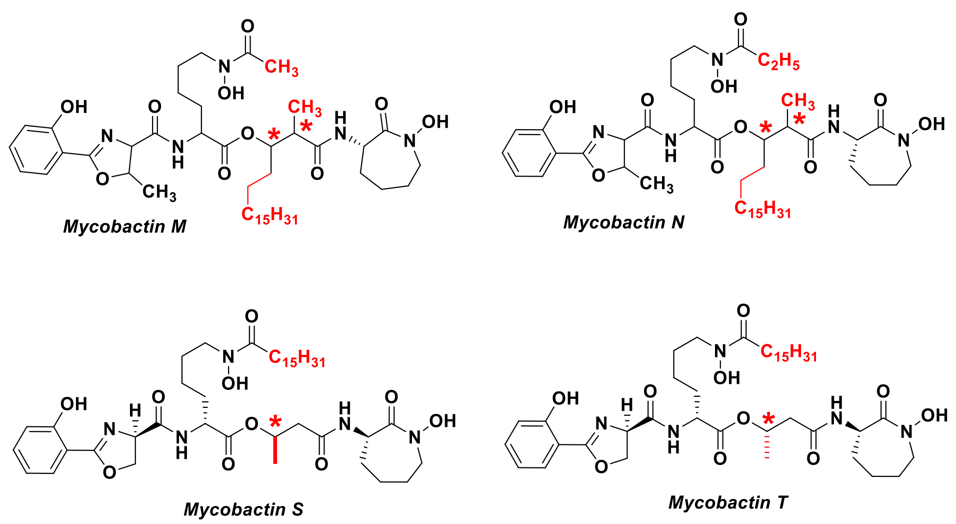

Figure 4. Chemical structure of the different mycobactin analogs, mycobactin M, mycobactin N, mycobactin S, and mycobactin T. Asterisk (*) indicates the chiral center in the respective mycobactin structures.

release into the intracellular environment, where it is taken up by membrane-bound lipophilic mycobactins. A study conducted on the exochelin-*Msmeg* scaffold indicates that the presence of the 3-hydroxamic acid group on the highly charged formylated pentapeptide backbone is responsible for Fe(III) chelation and formation of ferri-exochelin complex at physiological pH 5−6.8 from transferrin and ferritin.[82] (Figure 2)

Ferri-carboxymycobactin uptake occurs through a porin transport system, involving a facilitated diffusion process. Excess iron is transferred to the intra-envelope storage of iron, mycobactin. The ferri-reductase enzyme present at the intracellular membrane releases the Fe(II) from ferri-carboxymycobactin in the cytoplasm. Fe(II) forms a complex with salicylate residues and is transported to the mycobacterial iron storage apoprotein−bacterioferritin[83−85] (Figure 3). A study performed by Rodriguez and Smith[86] using the irtAB mutant strain in C57B/6 mice model and THP1-macrophage cell line indicated that irtAB gene mutant strains were unable to utilize ferri-carboxymycobactin efficiently. They further postulated that the IrtAB might be a transporter of Fe-carboxymycobactin and that the iron transportation process may be regulated by

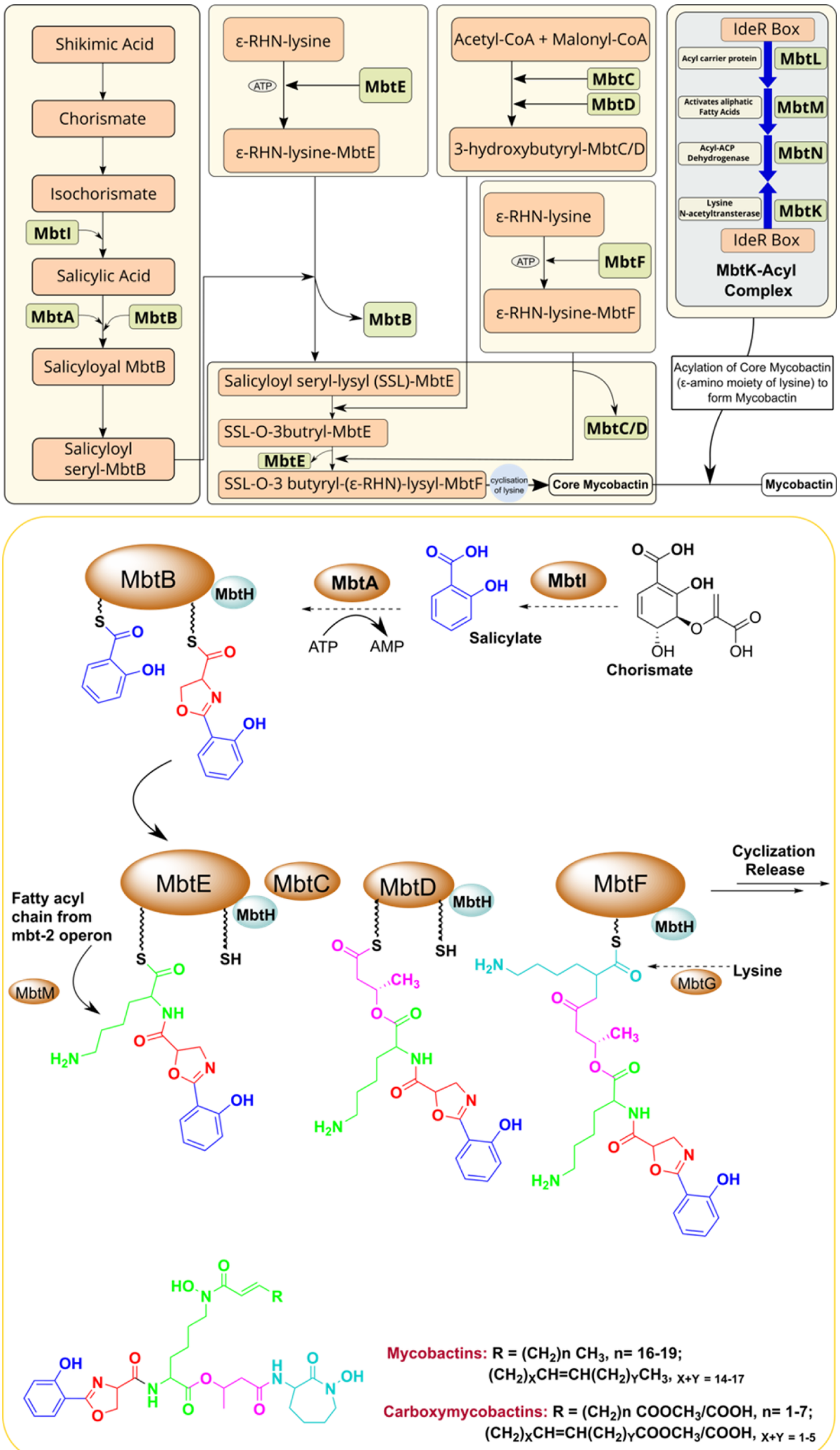

Figure 5. Schematic model of the mycobactin biosynthesis pathway. The biosynthesis pathway involves a complex of non-ribosomal peptide synthetase and polyketide synthase (NRPS-PKS) assembly chain. In this pathway, various building blocks, such as salicylate and modified lysine residues, are synthesized by the enzymes outside the assembly chains and covalently tethered to selected domains of the NRPS-PKS enzymes. Amino acids are subsequently added to the growing mycobactin molecule.

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the IdeR transcription factor. They concluded that preventing the Fe-carboxymycobactin uptake mechanism can hinder the survival of *mycobacteria* under iron-deficient conditions.

In the case of *Msmeg*, upon being recognized by the cell surface receptor protein (Rec) and, after that, via FxuD conjunction, the ferri-exochelin complex penetrates the membrane through the FxuA, FxuB, and FxuC system. The reductase enzyme present in the cytoplasm releases Fe(II) at the expense of ATP. Bacterioferritin present in the cytoplasm is responsible for the storage of Fe(II). In case of Rec and FxuABC transport system unavailability, the sequestration of Fe(II) from ferri-exochelin is carried out by mycobactin without energy input[87,88] (Figure 3).

The mycobactin biosynthesis pathway has long attracted interest as a viable source of anti-tubercular targets for the design and development of novel probes. Snow et al.[69] in 1970 first reported the experimental observation illustrating the inhibitory property of mycobactin M and N (isolated from *Mycobacterium marinum*) against *Mycobacterium paratuberculosis* and Mtb (Figure 4). This property was mainly attributed to their chemical structure that considerably differs from those of H37Rv mycobactin and carboxymycobactin. This finding prompted many researchers to synthesize analogs of mycobactins to study their ability to promote or inhibit mycobacterial growth.[89] The total synthesis of mycobactin S was achieved in 1983 by Maurer and Miller.[90] Mycobactin S (1997), an analog of mycobactin T (originally isolated by Snow et al. in 1965 from Mtb in their lab), reportedly had a different configuration at the chiral center on the 3-hydroxybutyrate carbon skeleton (Figure 4) and was found to inhibit the growth of Mtb by competitive iron chelation. However, Mtb revived due to the enhancement in the production of natural siderophores and their competitive iron chelation mechanism.[91] Poreddy et al.[92] made similar observations in 2003 and adopted a different approach targeting the Mtb iron acquisition pathway by investigating enzymes directly involved in the biosynthesis of mycobactins.[93]

Mycobactin biosynthesis is carried out by a mixed non-ribosomal peptide synthetase−polyketide synthase (NRPS-PKS) hybrid system encoded by 14 essential genes in the *mbt* gene cluster (*mbtA-mbtN*)[69,93−95] (Figure 5). The *mbt* gene cluster I (*mbtA-mbtJ*)[93] codes for enzymes (MbtA−J), responsible for the synthesis of the heterocyclic scaffold, i.e., 2-hydroxyphenyloxazolidine unit, whereas the addition of a terminal aliphatic residue to the salicyl-capped moiety is carried out by the MbtK−N enzymes encoded by the *mbt* gene cluster II (*mbtK-mbtN*).[93,96,97] The aromatic shikimate pathway[98,99] generates chorismate from shikimic acid, which acts as a starter unit for mycobactin biosynthesis. Next, MbtI catalyzes the transformation of chorismate to salicylic acid,[100−102] which is activated by the adenylating enzyme MbtA to the salicyl-adenosine monophosphate (Sal-AMP) at the expense of ATP. MbtA is a bifunctional enzyme that loads Sal-AMP onto the thiolation domain of MbtB, an acyl carrier protein, which is also part of the NRPS-PKS cluster. Vergnolle et al. reported that MbtA function is reversible and post-transnationally regulated by the acetylation mechanism in protein lysine acetyltransferase (Pat) and deacetyltransferase (DAc) enzymes.[103] Pat acetylates MbtA on residue Lys546, and the acetylation can be reversed by the NAD+-dependent DAc enzyme. The important connection between the Pat/DAc reversible acetylation system and MbtA together plays a significant role in Mtb survival at iron-limiting conditions for mycobacterial infection. In the macrophage phagosome model at low pH (pH 6) and iron-limiting conditions, ΔPat strains grow significantly faster than the wild type, while ΔDAc grows considerably slower. The study concluded that while acetylation of MbtA or FadD33 is prevented in ΔPat strains, resulting active enzymes can generate the required level of mycobactin for survival in the macrophage model. In contrast, in the ΔDAc strain, MbtA and FadD33 function at less than full potential, resulting in insufficient mycobactin production, which influences the Mtb survival of the macrophage. MbtB and MbtE are responsible for the inclusion of the serine and lysine residues, respectively, into the siderophore framework, followed by the addition of two malonyl CoA residues in the presence of MbtC and MbtD enzymes. The terminal lysine unit is incorporated by MbtF. Further modifications of the lysine residue are catalyzed by MbtK and MbtG, resulting in the development of the entire mycobactin scaffold.[104,105] MbtH plays a pivotal supporting role in the formation of soluble NRPS proteins, including MbtB, MbtE, and MbtF.[95] Additionally, the phosphopantetheinyl transferase (PPTase) enzyme plays a significant role by converting the *apo* carrier protein domains into the holo domains of mycobactin megasynthetase.[93,106] De Voss et al. performed MbtB gene replacement through homologous recombination modeling and with the hygromycin-resistance cassette of Mtb H37Rv, creating a mutant. The mutant strain showed restricted growth in iron-deficient medium but showed normal growth in the iron-supplemented medium. Moreover, the mutant failed to grow in the macrophage cell line as well as in THP-1 cells.[107] Chavadi et al.[94] (2011) reported that ΔmbtA, ΔmbtB, ΔmbtC, ΔmbtD, ΔmbtE, ΔmbtF, ΔmbtG, and ΔmbtT mutant *Msmeg* strains were not able to produce an appreciable amount of mycobactin, indicating the essentiality of Mbt ABCDEFGT NRPS-PKS enzymatic machinery in the biosynthesis of the core scaffold of the mycobactin. Bythrow et al.[108] constructed *Msmeg* Δ*EM* (deficient of both exochelin and mycobactin gene) which failed to show growth in an iron-limiting environment (GAS media, pH 6.6). Based on these findings, we can conclude that the absence of the mycobactin gene leads to growth inhibition in iron-deficient condition, thereby indicating the significance of the *mbt* gene. This evidence highlights the significance of the *mbt* gene cluster for survival in iron-deficient conditions.

To date, only four enzymes, i.e., MbtI, MbtA, MbtM, and PPTase, belonging to the mycobactin biosynthesis pathway, have been investigated as potential TB drug targets. This Perspective will focus on the small-molecule inhibitors reported so far, active against these four enzymes, e.g., MbtI, MbtA, MbtM, and PPTase. Based on reported inhibitors design and development efforts, a concise structure−activity relationship (SAR) discussion has also been provided. Moreover, the possibilities of exploring other enzymes as potential drug targets have also been discussed.[109] A short note on non-specific small-molecule development, targeting mycobactin biosynthesis pathway and non-conventional anti-tubercular drug delivery approach, i.e., the Trojan horse model in anti-tubercular drug discovery, has also been added.

2. DRUG DISCOVERY PROGRESS ON TARGETING ENZYMES INVOLVING MYCOBACTIN BIOSYNTHESIS PATHWAY

2.1. Salicylate Synthase (MbtI) Inhibitors. MbtI is a magnesium (Mg^{2+})-dependent bifunctional salicylate synthase

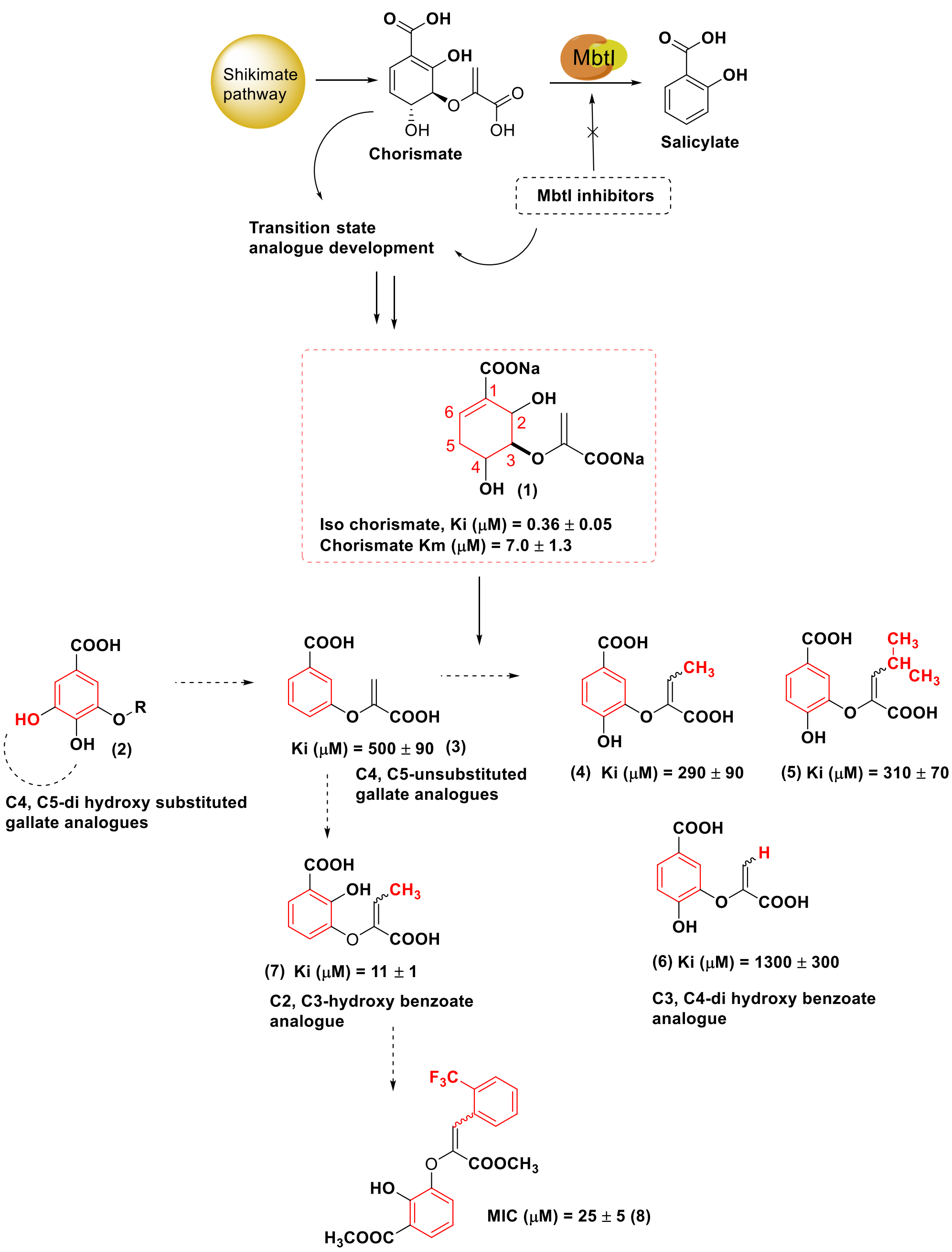

Figure 6. MbtI enzyme catalyzes the conversion of chorismate to salicylate. TS analogs **1**−**8** were developed based on chorismate structure to inhibit mycobactin biosynthesis in its early phases. The figure above depicts the journey from hit compound **1** to the optimized analog **7**, indicating structural modification with a significant MbtI inhibition profile. Other structural modifications **4**, **5**, and **6** led to negligible enzyme inhibition activity.

belonging to a large family of chorismate-utilizing enzymes (CUEs). The shikimate pathway plays a vital role in chorismate biosynthesis through the intermediate isochorismate.[102,110] MbtI catalyzes the conversion of chorismate to salicylate, which is further employed in the mycobactin assembly process. Apart from MbtI, other enzymes belonging to the CUEs family, such as 4-amino-4-deoxychorismate synthase (ADCS) and anthranilate synthase (AS), catalyze the production of folate and tryptophan, respectively. In addition to this, isochorismate synthase catalyzes the synthesis of both enterobactin and menaquinone.[111−113] Here, we have focused on MbtI due to its significance in salicylate production, which is the starting point of the mycobactin synthesis. Inhibition at this step effectively reduces the pathogenicity of Mtb in the iron-stress condition, without causing toxicity to the host, due to the absence of this enzyme (MbtI or its homologs) in the host cells. MbtI inhibitors are divided into two major classes based on chemical modifications, i.e., transition state (TS) analogs (Figures 6 and 7) and non-transition-state (non-TS) analogs (Figure 8). The core structure of a TS analog comprises a framework of chorismate, whereas non-TS analogs are small heterocyclic scaffolds-based inhibitors. These are discussed below in further detail.

2.1.1. TS Analogs. In 1991, Kozlowski and Bartlett[114] first designed and discovered chorismate TS analog (**1**) as a promising isochorismate synthase inhibitor, which produced a

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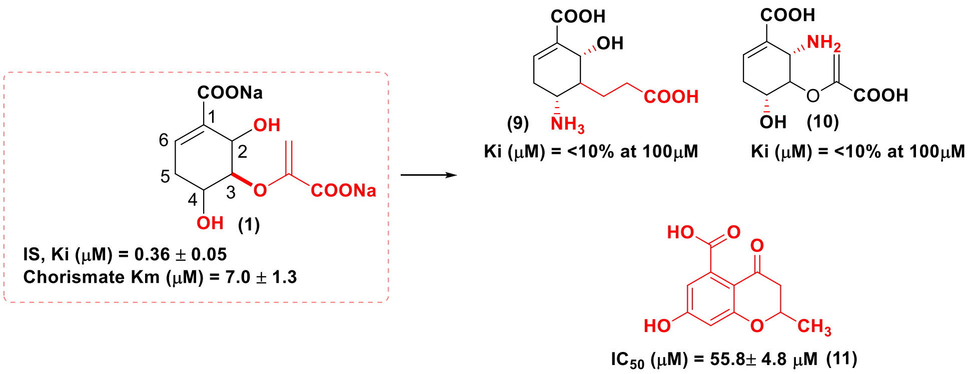


Figure 7. Chemical structure of MbtI inhibitors 9−11. The analogs 9 and 10 are based on hit scaffold 1, whereas the analog 11 is based upon ligand 7.

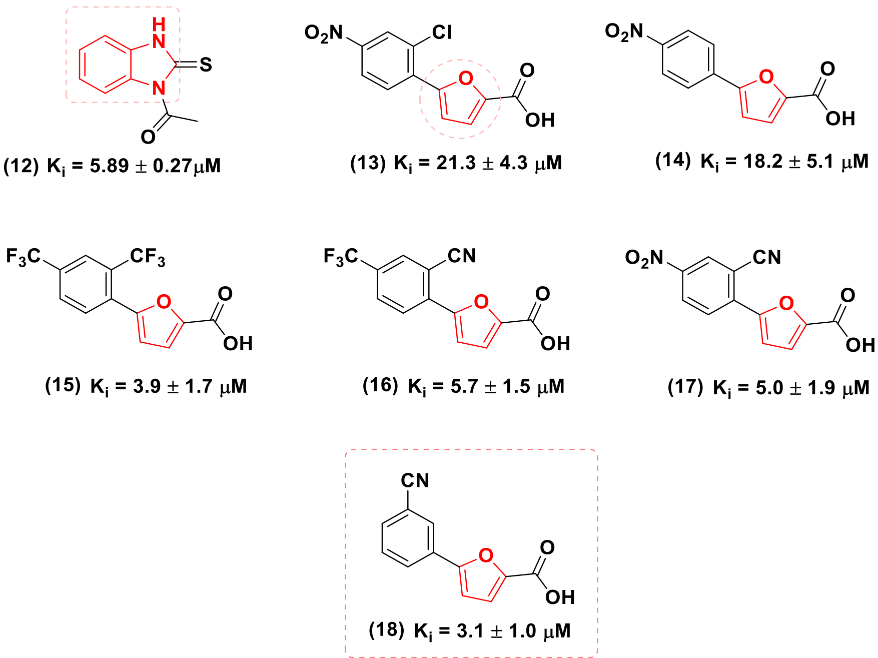


Figure 8. MbtI inhibitors development comprising the benzimidazole 12 and furan scaffolds 13−18 with a significant enzyme MbtI inhibition profile.

K_i value of 0.36 ± 0.05 μM in iso-chorismate inhibition study. This discovery further stimulated a search for more potent analogs with the same mechanism of action. Manos-Turvey et al.[115] in 2010 synthesized several novel TS analogs based on the gallate scaffold. Modification of the gallate framework, including a dihydroxy substitution on the aromatic ring at the C-4 and C-5 positions (**2**), resulted in derivatives with inferior biological activity. In contrast, the analog devoid of C-4 and C-5 hydroxyl groups at the aromatic ring (**3**) exhibited better enzyme inhibitory activity and anti-bacterial properties. Subsequently, studies shifted toward the exploration of 3,4-dihydroxybenzoic acid-based inhibitors. Substitution with small aliphatic groups, such as methyl and isopropyl units, at the C-3 position of the carboxyvinyl residue, yielded better analogs (**4**, **5**) compared to the C3-enol-pyruvyl-substituted derivative (**6**) in terms of MbtI enzyme inhibitory activity. A moderate improvement in the enzyme inhibition profile was observed in the case of enol-pyruvate linkage substitution at the C-2 and C-3 positions of the aromatic ring. Among all the designed analogs, the compound with a 2,3-dihydroxybenzoate scaffold bearing the but-2-enoate (**7**) substitution at the C-3 position showed the most promising inhibitory activity, with a K_i value of 11 ± 1 μM against MbtI. However, compound **7** exhibited poor anti-tubercular activity in whole-cell screens against Mtb, giving an MIC_{50} value >1 mM reported in the study performed by the same research group.[116] Further to this, the same research group designed and evaluated a series of 3-phenylacrylate analogs as MbtI inhibitors. None of the compounds synthesized in this attempt showed improved inhibition profile against MbtI compared to **7**, although analog (**8**) has emerged with a significant *in vitro* anti-tubercular activity with an MIC_{50} value of 25 ± 5 μM in iron-supplemented media[116] (Figure 6).

In 2015, Liu et al.[117] designed TS analog (**9**) based on the parent scaffold **1**. However, this compound exhibited suboptimal MbtI inhibition activity (i.e., <10%) at a concentration of 100 μM. The introduction of a CH_2 moiety as a bioisosteric replacement for the C-5 oxygen atom of chorismate resulted in the loss of a critical hydrogen bond with the Arg405 residue in the active site of MbtI enzyme, yielding an inactive analog. Also, introducing a protonated amino group at the C-4 position of the aromatic ring of **9** resulted in a significant repulsive electrostatic interaction with the Arg405 residue, which made an additional negative impact on the enzyme MbtI inhibition activity (Figure 7).

In 2017, Zhang et al.[118] designed TS analog (**10**), within which the C2-OH of the parent scaffold **1** was replaced with an amino group. However, analog **10** also failed to exhibit any significant enzyme inhibitory activity, K_i < 10% at 100 μM concentration (Figure 7).

In 2018, Pini et al.[119] replaced the phenyl ring of compound **7** with a chromane framework and designed a library of novel MbtI inhibitors bearing the benzodihydropyran scaffold. Analog (**11**) exhibited an IC_{50} value of 55.8 ± 4.2 μM in the MbtI enzyme inhibition assay and emerged as the most potent one among the other library members (Figure 7).

2.1.2. Non-TS Analogs. In 2010, Vasan et al.[120] designed novel benzisothiazolones, diaryl sulfones, and benzimidazole-2-thione class of compounds as non-TS analogs. The ligand benzimidazole-2-thione (**12**) emerged as the most promising MbtI inhibitor, with a K_i value of 5.89 ± 0.27 μM in high-throughput enzyme inhibition assays. In 2018, Chiarelli et al.[121] synthesized a series of furan ring-containing MbtI inhibitors based on hits identified through *in silico* screening of a commercially available library against the crystal structure of MbtI (PDB ID: 3VEH).[122] Hit compound (**13**), a 2-phenylfuran derivative, was tested *in vitro* in MbtI inhibition assay and showed a K_i value of 21.1 ± 4.1 μM against enzyme MbtI. Therefore, compound **13** was considered as the starting point for further development. SAR investigations into the structure of **13** led to the synthesis of analog **14**, which carried only one structural difference, i.e., removal of the *o*-chloro group at the aromatic ring, and showed MbtI inhibition activity with K_i of 5.3 μM, albeit with low anti-tubercular activity (MIC_{99} = 156 μM) against Mbt H37Rv in a whole-cell assay (Figure 8). Further, the same research group in 2019[123] reported a small library of 16 furanic acid analogs with MbtI inhibitory activity. Analog (**15**), with *ortho*,*para*-di-CF_3 substitution on the aromatic ring, was the most potent candidate, with a K_i value of 3.9 ± 1.7 μM against MbtI inhibition assay and IC_{50} of 13.1 ± 2.0 μM in a whole-cell assay against Mtb H37Rv. Another significant outcome of this study was the discovery of two other derivatives, (**16**) and (**17**), with an *ortho*-CN group substitution on the aromatic ring. The only structural difference between the two compounds was a *para* substitution on the aromatic ring. The candidate with additional *p*-CF_3 substitution **16** emerged as slightly better analog (K_i = 5.7 ± 1.5 μM, IC_{50} 18.5 ± 3.2 μM) compared to the candidate with *p*-NO_2 substitution **17** (K_i = 5.0 ± 1.9 μM, IC_{50} 24.4 ± 5.9 μM). The same group in 2020[124] reported a pilot library of furan analogs with different substitutions at the *meta*-position on the aromatic ring. The study revealed that analog (**18**), with a *m*-CN-group-substituted aromatic group, showed the most promising activity, with a K_i value of 3.1 ± 1.0 μM in the MbtI inhibition assay and IC_{50} of 6.3 ± 0.9 μM in a whole-cell assay against Mtb H37Rv. Analog **18** emerged as the most potent competitive inhibitor of MbtI known to date.

2.2. Salicyl-AMP Ligase (MbtA) Inhibitors. MbtA-targeted anti-tubercular agents so far constitute the dominant class of mycobactin biosynthesis inhibitors. These types of molecules have been designed to target the initial step of the mycobacterial siderophore production catalyzed by MbtA, i.e., activation of salicylate with the formation of salicyl-adenosine monophosphate (Sal-AMP) intermediate (**19**). MbtA assembles the mycobactin peptide core and acts as a bifunctional enzyme that converts salicylate into Sal-AMP and loads the latter intermediate on to the phosphopantetheinylation domain of MbtB.[107] The majority of MbtA inhibitors are nucleobase-modified adenosine analogs that mimic the Sal-AMP structure and block the enzyme activity. However, several non-nucleoside MbtA inhibitors[125] are currently being developed. The archetypal MbtA inhibitor scaffold comprises an aromatic ring, a linker bridge, a sugar unit, and a base moiety. All these structural modifications have been summarized with respect to newer inhibitors' development in the sections below.

2.2.1. Linker Modification. Ferreras et al. (2005)[126] reported the design and synthesis of 5′-*O*-*N*-salicylsulfamoyl adenosine (Sal-AMS) (**20**), a congener of Sal-AMP **19**, as a MbtA inhibitor (Figure 9). The molecular framework of ligand **20** resembled that of the naturally occurring intermediate Sal-AMP **19**, and the only structural difference was constituted by the sulfonamide linkage replacing the orthophosphate ester group. Sal-AMS **20** exhibited significant MbtA inhibitory activity, with a K_i value of 10.7 ± 2.0 nM, and emerged as the first biochemically confirmed inhibitor of siderophore biosynthesis. Sal-AMS showed 18-fold higher anti-tubercular activity (MIC_{50} value of 2.2 ± 0.3 μM) against Mtb H37Rv strains, cultured in iron-deprived conditions, compared to bacteria grown in iron-rich media (MIC_{50} value of 39.9 ± 7.6 μM). In 2013, Lun et al.[127] studied ligand **20** in a TB-infected acute murine model over 4 weeks (intraperitoneal administration, 5.6 or 16.7 mg/kg dose) and reported that Sal-AMS also did not exhibit any *in vitro* cytotoxicity against P388, HepG2, and Vero cell lines (CC_{50} > 200 μM). In contrast, the pharmacokinetic (PK) profile of **20** showed that this ligand had poor oral bioavailability (F < 2%), low clearance (Cl = 4.9 mL min^{-1} kg^{-1}), low microsomal stability ($t_{1/2}$ > 30 min), and a small volume of distribution (V_d = 0.079 L kg^{-1}). Due to its suboptimal PK profile, **20** was not taken forward in clinical trials, but the initial *in vitro* and *in vivo* investigations on its efficacy and toxicity profile inspired subsequent drug discovery campaigns by other research groups aiming to develop novel Sal-AMS analogs with more favorable metabolic properties and improved PK profile.

Somu et al. in 2006[128] reported the synthesis of Sal-AMS analogs with several bioisosteric replacements of the cleavable acyl phosphate linker of **19** and modifications of the salicyl moiety. The compounds were evaluated for their ability to inhibit the growth of Mtb H37Rv under iron-deprived conditions and disrupting the activity of the adenylating enzyme MbtA. These rationally designed MbtA inhibitor nucleosides contained three types of acyl phosphate moiety-mimicking linkages, including acylsulfamate and acylsulfamide β-ketophosphonate, and acyltriazole linkers. Nucleosides containing the β-ketophosphonate and acyltriazole tether structural modification failed to exhibit any efficacy against H37Rv at 100 μM concentration. However, the acylsulfamide linker-bearing analog (**21**), obtained by substituting the oxygen atom of the sulfamate group with a nitrogen atom, showed notable activity in whole-cell assays, with an MIC_{99} value of 0.19 μM, similar to that of isoniazid (MIC_{99} = 0.18 μM), and an MIC_{50} value of 0.077 ± 0.022 μM under iron-deficient medium.

Analog **21** was assessed for mycobactin biosynthesis inhibitory activity using a modified radioassay method, adapted

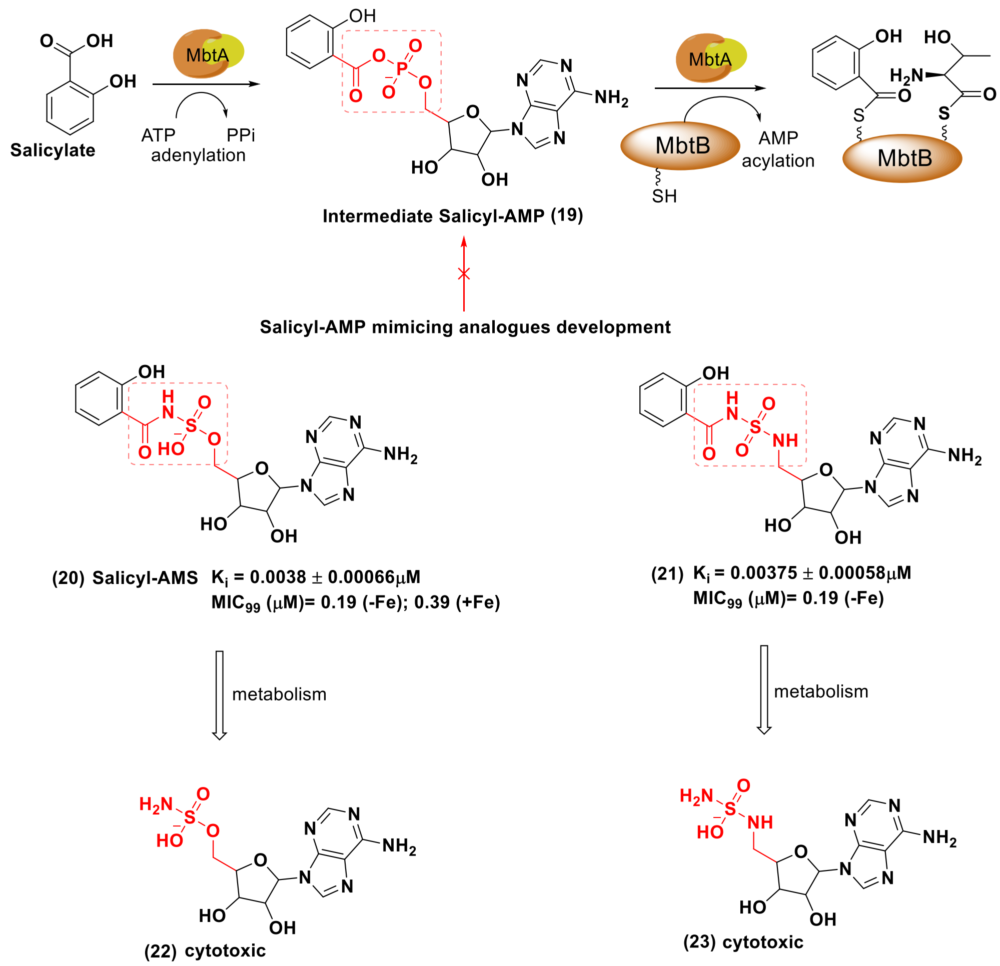


Figure 9. MbtA is a bifunctional enzyme that catalyzes the conversion of salicylate to Sal-AMP **19** and the loading of **19** onto the MbtB thiolation domain. The first MbtA inhibitor developed to target the Sal-AMP biosynthetic step was Sal-AMS **20**. This lead compound **20** exhibited a remarkable anti-tubercular activity in iron-efficient (+Fe) and iron-deprived (−Fe) medium, although its PK profile was suboptimal. This prompted further studies to discover analogs with more favorable PK parameters and drug-like properties, such as nucleoside **21**, without compromising their Mtb H37Rv-growth inhibitory activity. Hydrolysis of the acylsulfamide linkage of derivatives **20** and **21** resulted in cytotoxic metabolites **22** and **23**.

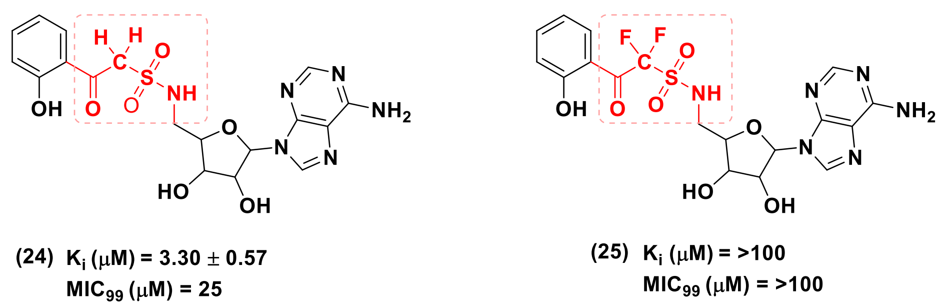
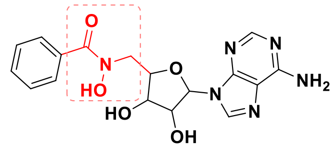


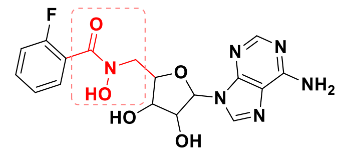
Figure 10. Chemical structures of analogs **24** and **25** with respective MbtA inhibition values and anti-tubercular profile in iron-deprived GAST medium. The major structural modification carried out with the *β*-ketosulfonamide linker resulted in poor candidate development.

from the one reported by DeVoss et al.[107] In this comparative study, complete inhibition of mycobactin and carboxymycobactin biosynthesis was observed at a concentration of 10 *μ*M. In 2006, Vannada et al.[129] further evaluated the MbtA inhibition activity of nucleoside **21** using a [^{32}P] PPi-ATP exchange assay and showed that the ligand **21** was a potent MbtA inhibitor with a K_i value of 0.0038 ± 0.0006 *μ*M (Figure 9). Furthermore, hydrolysis of the acylsulfamide linkage of derivatives **20** and **21** led to the production of cytotoxic metabolites (**22** and **23**), and ionization of the sulfamate linker at physiological pH resulted in limited oral bioavailability (Figure 9).

Vannada et al.[129] further modified the linker unit of the anti-tubercular nucleosides by substituting the labile acylsulfamate moiety of **21** with the more chemically stable *β*-ketosulfonamide linker (Figure 10). However, this structural modification yielded a nucleoside (**24**) and fluorinated analog (**25**) with poor drug-like properties, with K_i values of 3.30 ± 0.57 and >100 *μ*M against MbtA, respectively, compared to the parent

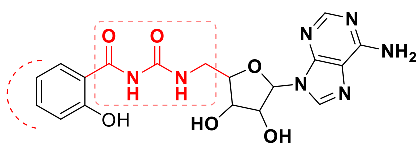


(26) DhbE inhibition 1.8% at 12.5μM concentration

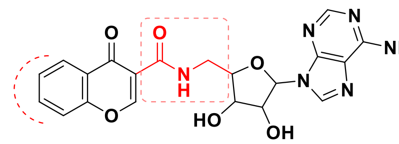


(27) DhbE inhibition 0.1% at 12.5μM concentration

Figure 11. Chemical structures and enzyme DhbE inhibition profile of BEN-AMN **26** and F-BEN-AMN **27**.



(28) K_i (μM)= 1.25 ± 0.05
MIC_{99} (μM)= 200 (-Fe); >200 (+Fe)



(29) K_i (μM)= >100
MIC_{99} (μM)= 12.5-25 (-Fe); 50 (+Fe)

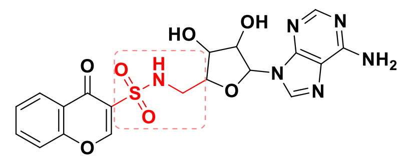
Figure 12. Chemical structures and anti-tubercular profile of analogs **28** and **29** with diamide and amide linkage, respectively.

acylsulfamide derivative **21**. This approach has not been taken further into consideration for analog development.[129]

In 2006, Miethke et al.[130] designed and synthesized several arylsulfamoyl adenosine analogs of Sal-AMS and tested their inhibitory activity against 2,3-dihydroxybenzoate-AMP ligase (enterobactin synthetase component E, or DhbE). This enzyme is involved in the synthesis of DHB-capped siderophores, termed bacillibactins, produced by *Vibrio* spp, *Bacillus* spp, and other enteric bacteria. Sal-AMS **20** exhibits inhibitory activity against this enzyme. The compounds were tested at a concentration of 12.5 μM, and the nucleosides containing hydroxamoyl linkage (BEN-AMN, **26**) showed only 1.8% of DhbE inhibition compared to Sal-AMS, which showed 99.6% inhibition. This weak activity was attributed to the shortening of the hydroxamoyl linker leading to impairment of the fitting of the aryl or adenosine moiety to the DhbE active site. Notably, a derivative bearing the dihydroxybenzoylsulfamoyl linkage (DHB-AMS) inhibited the enzyme marginally better than Sal-AMS **16**, with 99.8% inhibition. Further introduction of a fluorine atom in the aromatic ring resulted in an inactive analog F-BEN-AMN (**27**) that showed only 0.1% of DhbE inhibition at 12.5 μM concentration. Further research in this direction was ceased (Figure 11).

In 2007, Qiao et al.[131] first suggested the introduction of the urea linker to connect the adenosine unit with the *o*-hydroxyphenyl ring. This resulted in acyclic acyl urea derivative (**28**) and nucleoside (**29**), in which a chromane scaffold was linked via an amide bond to the sulfamoyl unit of Sal-AMS (Figure 12). Analog **28** showed a 329-fold loss of anti-tubercular potency (K_i = 1.25 ± 0.05 μM) compared to Sal-AMS **20**. Analog **29** also exhibited poor activity, with a K_i value of >100 μM against MbtA.

In 2013, Engelhart and Aldrich[132] designed four conformationally constrained analogs of Sal-AMS **20** with linker modification by introducing chromone, quinolone, benzoxazinone-3-sulfonamide, and 8-hydroxy chromane units into the framework of parent compound **20** to remove two rotatable bonds and the ionizable sulfamate group. This synthetic strategy is aligned with the observations by Veber et al.,[133] who suggested that reduced molecular flexibility of drug candidates can be an important predictor of good oral bioavailability. The sulfonamide linkage of adenosine with the chromane (Figure 13) scaffold yielded nucleoside (**30**), which showed non-



(30) Ki^{app} (μM)= 0.12 ± 0.02
MIC_{99} (μM)= >50 (-Fe); >50 (+Fe)

Figure 13. Chemical structure and anti-tubercular profile of conformationally constrained nucleoside analog **30**. Two major structural modifications have been considered: (a) introduction of constrained sulfonamide linkage and (b) replacement of the aromatic ring with chromane moiety. All these modifications resulted in poor candidate development.

promising activity against Mtb H37Rv in both iron-deficient conditions (MIC_{99} > 50 μM) and iron-rich media (MIC_{99} > 50 μM). The lack of an ionizable functional group in the nucleoside derivative **30** led to a significant reduction in inhibitory activity against the MbtA enzyme, with a K_i value of 0.12 ± 0.02 μM. This linker modification was not taken into account further as part of favorable design efforts in subsequent studies.

2.2.2. Aromatic Group Modification. The *ortho*-hydroxyl group within the salicylate residue is another important structural feature necessary for the biological and biochemical activity of Sal-AMS analogs. The different design approaches indicate that removal or substitution of the hydroxyl group with another group considerably affects the enzymatic inhibitory activity and anti-tubercular properties when compared to parent ligand Sal-AMS **20**. Besides the *ortho* position, substitution at other positions, i.e., *meta* or *para* or disubstitution approaches, should also be accounted for in newer candidate development purposes. In their early work, Somu et al.[128] observed that Sal-AMS analogs containing a 2-amino benzoate residue (**31**) or an unsubstituted benzene ring (**32**) showed complete ablation (MIC_{99} > 100 μM) or significant reduction (MIC_{99} = 12.5 μM), respectively, of Mtb H37Rv growth inhibitory properties (Figure 14).

Following these studies, Qiao et al.[131] in 2007 further explored the importance of substitution at the C-2 position of the salicyl moiety. These authors found that the C2-hydroxy

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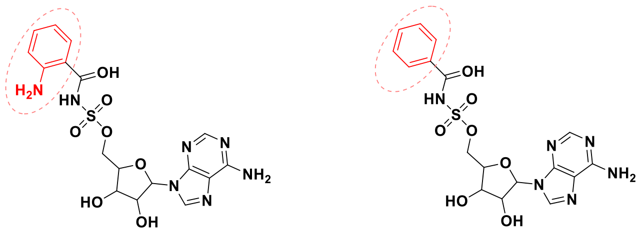


Figure 14. Chemical structures and anti-tubercular profile of analogs **31** and **32**.

group was essential for anti-tubercular efficacy. Fluoro substitution at the C-4 position of the aromatic ring yielded compound (**33**), with significant growth inhibitory activity in whole-cell screening against Mtb H37Rv under iron-limiting conditions (MIC_{99} = 0.098 μM) compared to iron-rich conditions (MIC_{99} = 0.39 μM). Analog **33** exhibited a significant MbtA inhibition activity, with a K_i value of 0.012 ± 0.0016 μM. Derivative (**34**), bearing an additional C4-amino group on the aromatic ring, showed improved enzyme inhibition activity (K_i = 0.040 ± 0.004 μM) and promising selectivity against Mtb grown in the iron-deprived medium compared to Mtb cultured in an iron-rich environment, with MIC_{99} values of 1.56 and 25 μM, respectively. Nelson et al.[134] performed PK profile studies on **34**. The poor AUC_{po} and C_{max} data gathered for this compound indicated that the presence of the *para*-amino group enhanced polarity and led to a faster clearance (Figure 15).

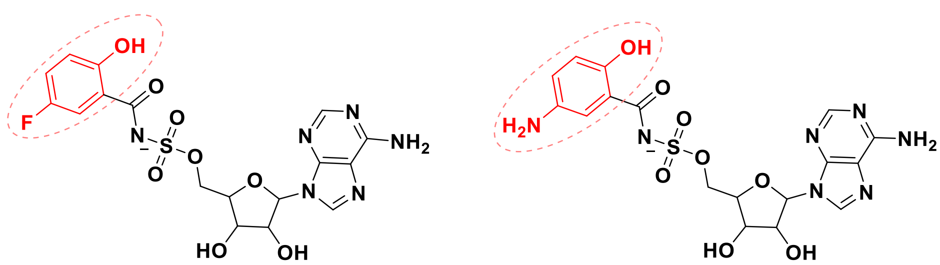
In contrast, substitutions of the salicyl moiety with morpholine (**35**), cycloalkyl (**36** and **37**), alkyl (**38**), alkyloxy (**39**), methoxymethane (**40**), and aminoacyl (**41**) units were poorly tolerated, yielding nucleoside derivatives weakly active against the tubercle bacillus H37Rv (Figure 16).[131] The Qiao research group also designed and developed some pyridyl analogs (**42** and **43**) with different substitutions at the second position of the heterocyclic ring, yielding a derivative containing a tetrafluoro 4-azido benzene moiety (**44**). However, this synthetic approach failed to elicit the desired inhibitory effects in both whole-cell screening (MIC_{99} > 200 μM against Mtb in iron-deficient and replenished conditions) and enzymatic assay levels. As an exception, the pyridine analog (**45**) with chlorine substitution at the second position of the heterocyclic ring showed moderate anti-tubercular and MbtA inhibition activity. Compound **45** can be further considered as the lead for the development of better candidates (Figure 16).

Engelhart et al.[132] developed conformationally constrained Sal-AMS analogs. Design strategy involved incorporating constrained linkers tethering bulkier heterocyclic rings, such as chromone, quinolone, and benzoxazinone-3-sulfonamide units, as the replacement of the *o*-hydroxy aromatic ring. Parent chromane analog **30** has already been discussed in the Sal-AMS linker modification section of this Perspective. The constrained linker−nucleosides series designed by Engelhart et al. included analogs bearing 8-hydroxychromone (**46**), benzoxazinone-3-sulfonamide (**47**), and quinolone (**48**) moieties. Quinolone-including nucleoside **48** showed a MbtA enzyme inhibitory activity with a K_i value of 120 nM. This activity can be attributed to the ionizable nitrogen proton of the heterocyclic ring (pK_a = 7.8 circa). However, this compound **48** was not active against Mtb in whole-cell screens. At this stage, it became apparent that the formal negative charge carried by the sulfamate nitrogen of Sal-AMS **20** was necessary for binding the MbtA active pockets and that a pK_a value closer to that of the parent compound (pK_a of **20** = 3 circa) was key to successful membrane permeability (Figure 17).

These challenges encountered in the development of optimal nucleoside MbtA inhibitors were addressed by Dawadi et al.,[135] who designed other conformationally constrained Sal-AMS analogs using the cinnolone scaffold (**49**). Cinnolone-bearing analogs showed superior anti-tubercular profile and drug-like properties compared to benzoxazine **47**- and quinolone **48**-containing compounds. The halogenated 7-fluorocinnolone analog (**50**), which had a pK_a of 6.3, showed the most promising anti-tubercular activity (MIC = 2.3 μM) under iron-deficient conditions among the designed derivatives and inhibited MbtA with a K_i of 12 nM (Figure 18). PK profile analysis of **50** indicated a 7-fold higher volume of distribution in the female Sprague−Dawley rat model, resulting in a 2-fold improvement of half-life compared to Sal-AMS **20**. The results were encouraging and favored the use of a cinnolone moiety as a bioisosteric replacement of salicyl-sulfamate to provide MbtA inhibitor nucleosides with an improved PK profile.

2.2.3. Sugar Moiety Modification. The structural modification of the sugar moiety, combined with alterations at the linker and aromatic nucleus sites, played a crucial role in developing Sal-AMS **20** analogs with enhanced MbtA inhibition potency and anti-tubercular properties.

Early SAR[136] explorations of the glycosyl moiety of Sal-AMS **20** revealed that deletion of hydroxyl groups in the ribose sugar and replacement of the sugar moiety with a pentane ring was well tolerated. Derivatives containing a carbocyclic unit (**51**)



(**33**) K_i (μM) = 0.012 ± 0.0016
MIC_{99} (μM) = 0.098 (-Fe); 0.39 (+Fe)

(**34**) K_i (μM) = 0.040 ± 0.004
MIC_{99} (μM) = 1.56 (-Fe); 25 (+Fe)

Figure 15. Chemical structures and anti-tubercular profile of analogs **33** and **34**. Disubstituted aromatic scaffold-comprising analogs emerged as promising candidates.

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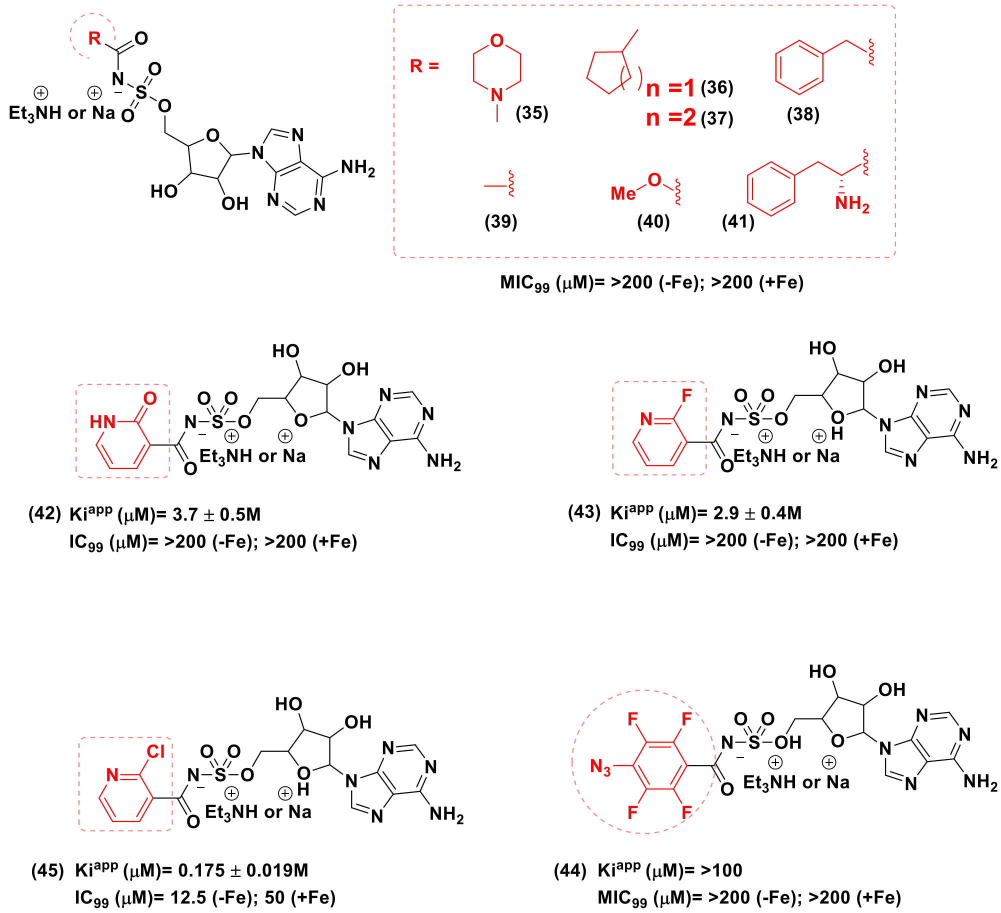

Figure 16. Chemical structure and anti-tubercular profile of analogs **35**−**45**.

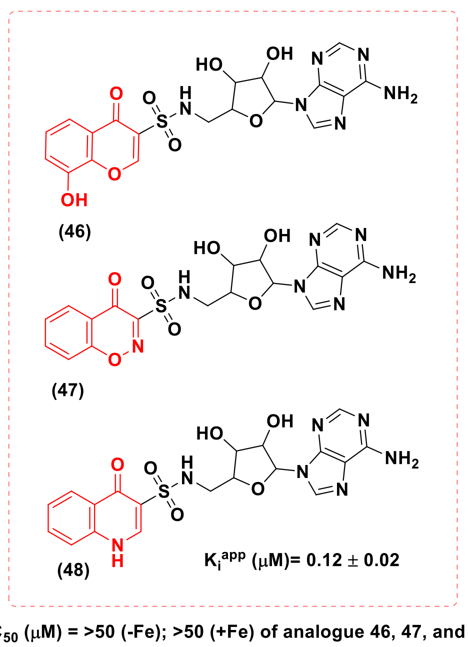

Figure 17. Chronological development order of bulky aromatic substituted nucleosides comprising chromane **46**, benzoxazinone-3-sulfonamide **47**, and quinolone **48** groups, respectively. Analogs **46**, **47**, and **48** were synthesized to address the poor PK profile of the parent compound **20**. Among the series, quinolone analog **48** showed a significant enzyme MbtA inhibition profile.

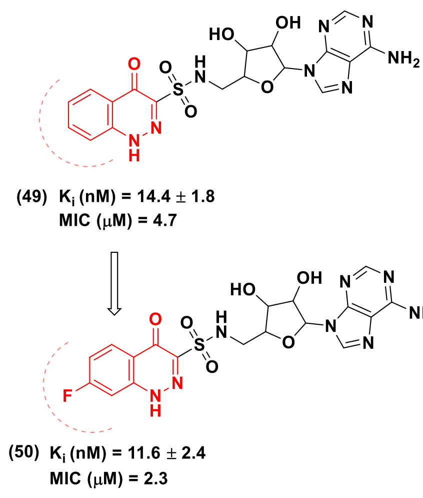

Figure 18. Chromone, quinolone, and cinnolone scaffolds provided a suitable alternative to the *ortho*-hydroxybenzoate unit and corresponding Sal-AMS **20**. The fluorinate cinnolone-containing ligand **50** was found to retain the essential anti-tubercular activity and MbtA inhibitory property of **20** with an improved PK profile.

and 3′-deoxyribose unit (**52**) exhibited excellent anti-tubercular activity in the whole-cell assay against Mtb H37Rv in iron-deprived conditions. Both compounds had a MIC_{99} value of 1.56 μM, which represented a 5-fold decrease in activity compared to **20**. These two nucleoside derivatives **51** and **52** also displayed potent MbtA inhibition with a 2−3-fold

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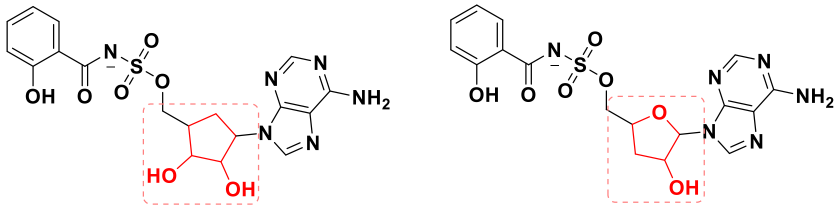
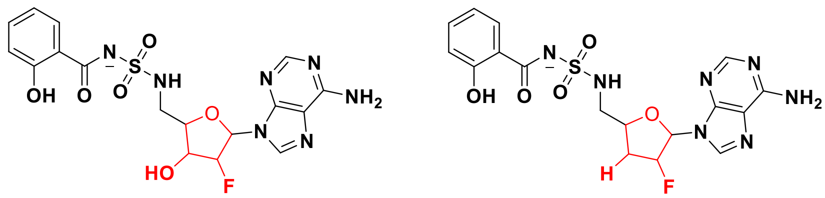


(51) K_i = 2.3 nM
MIC_{99} = 1.56 μM

(52) K_i = 3.3 nM
MIC_{99} = 1.56 μM

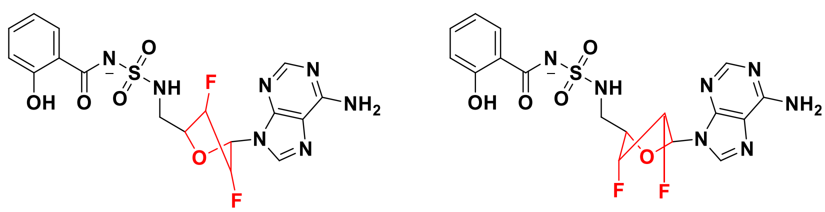
Figure 19. Chemical structure and anti-tubercular profile of analogs **51** and **52**.



(53) K_i = 1.7 ± 0.1 nM
MIC = 0.19 μM

(54) K_i = 3.0 ± 0.2 nM
MIC = 0.78 μM

Figure 20. Chemical structures and anti-tubercular profiles of analogs **53** and **54**. Analogs with fluoro-substituted sugar moiety showed favorable anti-tubercular and MbtA inhibition properties and an improved PK profile compared to **21**.



(55) K_i = 1.4 ± 0.3 nM
MIC = 0.78 μM

(56) K_i = 139 ± 10 nM
MIC = 37 μM

Figure 21. Strategies developed by the Aldrich group toward the modification of the sugar moiety of MbtA nucleosides. Adding two fluorine atoms at positions 2′ and 3′ of the sugar ring resulted in different configurations, e.g., a 3′-*endo*, 2′-*exo* (North, **55**) or a 2′-*endo*, 3′-*exo* (South, **56**), leading to a significant improvement of pharmacokinetic profile.

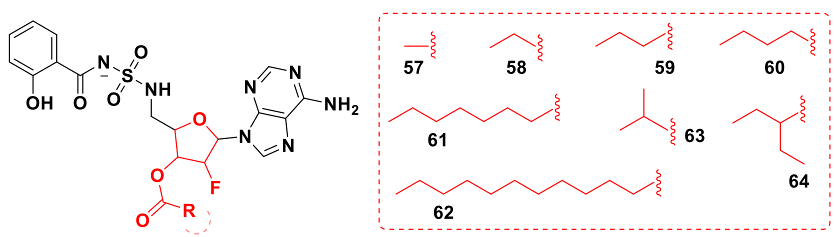
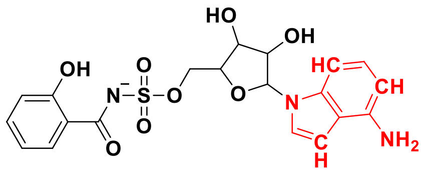


Figure 22. Chemical structure of the ester analogs **57**−**64** of parent compound **53**.

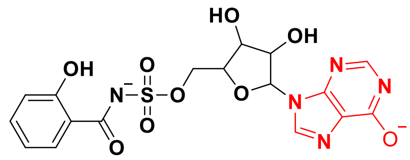
increase in activity compared to Sal-AMS, with K_i values of 2.3 and 3.3 nM, respectively (Figure 19).

Nelson et al. (2015)[134] reported ligands which contained a fluorinated ribose sugar unit and the acylsulfamide linker of **21**, including derivatives (**53**) and (**54**) with improved PK profiles and MbtA enzyme inhibitory activity. To address the potential shortcomings of **21**, Nelson and co-workers synthesized several analogs of this compound to improve its poor drug-like properties by modifying their physicochemical properties, such as acidity and lipophilicity. The authors also performed a complete *in vivo* PK evaluation of the compounds in the Sprague−Dawley rat model. Replacement of the 2′- hydroxyl group with fluorine gave compound **53**, and additional removal of the 3′-hydroxyl group in the ribose unit afforded analog **54**. Compounds **53** and **54** showed a 3-fold improvement in oral exposure with a 3- and 6-fold increase in half-life, respectively, compared to parent ligand Sal-AMS. Replacement of the 2′-hydroxyl group with a fluorine atom resulted in a dramatic improvement in the half-life that translated to the improved oral exposure (Figure 20).

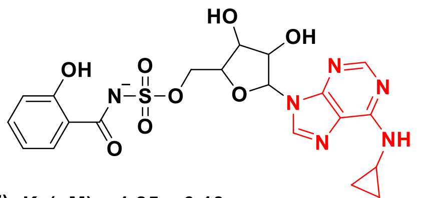
Dawadi et al.[137] further explored the contribution of the fluorinated ribose unit to the anti-tubercular activity, enzyme inhibition efficacy, and drug disposition profile of the MbtA nucleoside inhibitors by incorporating fluorine atoms at both 2′ and 3′ positions of the glycosyl unit. Incorporation of fluorine profoundly affected the stereoelectronic properties of the nucleosides' ribose moiety, orienting the sugar ring into either a 2′-*endo*, 3′-*exo* (South), or a 3′-*endo*, 2′-*exo* (North)



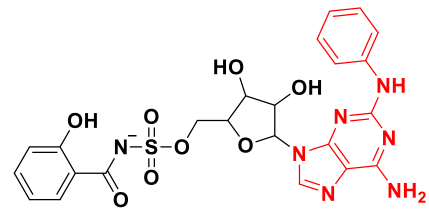
(65) K_i (nM) = 20.1 ± 2.3
MIC_{99} (μM) = 6.25 (-Fe); 50 (+Fe)



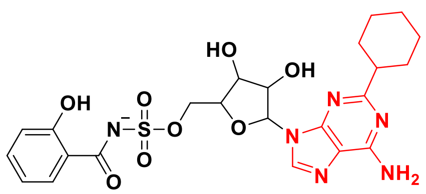
(66) K_i (nM) = 800 ± 50
MIC_{99} (μM) = >100 (-Fe); >100 (+Fe)



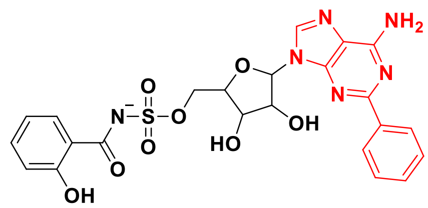
(67) K_i (nM) = 1.85 ± 0.13
MIC_{99} (μM)= 0.098 (-Fe); 6.25 (+Fe)



(68) K_i (nM) = 0.94 ± 0.16
MIC_{99} (μM) = 0.049 (-Fe); 0.39 (+Fe)



(69) K_i (nM) = 0.40 ± 0.05
MIC_{99} (μM) = 0.049 (-Fe); 0.39 (+Fe)



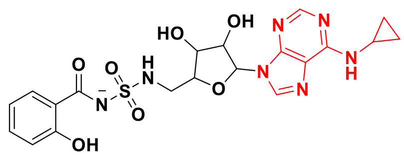
(70) K_i (nM) = 0.27 ± 0.07
MIC_{99} (μM) = 0.049 (-Fe); 0.39 (+Fe)

Figure 23. Base modification of Sal-AMS scaffold resulted in some novel biologically efficient analogs, **65**−**70**.

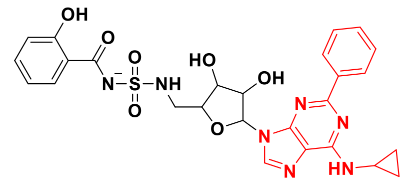
conformation (Figure 21). Compound (**55**), which had the ribose unit in the North (C3′-*endo*) conformation, showed notable activity in the biochemical and whole-cell assays (K_i = 1.4 nM, MIC = 0.78 μM), whereas derivative (**56**) with the South (C2′-*endo*) conformation showed an improved PK profile as evaluated using an *in vivo* cannulated rat model. Derivative **56** showed a 4-fold increase in bioavailability (F% 5.3 ± 1.3), a 15-fold improvement in C_{max}, a 75-fold increase in oral AUC, and a 25-fold greater half-life compared to the lead analog **20**.

In 2016, the Dawadi group[138] sought to improve the bioavailability of promising derivative **53** and used a prodrug approach to protect the 3′-hydroxyl unit from detrimental pre-systematic hydrolysis by intestinal esterases. Different linear and branched alkanoyl groups, such as methyl (**57**), ethyl (**58**), *n*-propyl (**59**), *n*-butyl (**60**), *n*-heptyl (**61**), *n*-undecyl (**62**), isopropyl (**63**), and 3-pentyl (**64**) ester linkages, were incorporated at the 3′ position of the 2′-fluorinated analog **53** to produce a series of lipophilic ester prodrugs (Figure 22). Although the prodrugs were stable in mouse, rat, and human serum, the compounds exhibited reduced permeability (apical-to-basolateral direction) in the Caco-2 cell transwell model, with 5−28 times higher efflux ratio compared to **53**. This prodrug strategy did not yield the desired results, as it led to a lower oral availability than **53**.

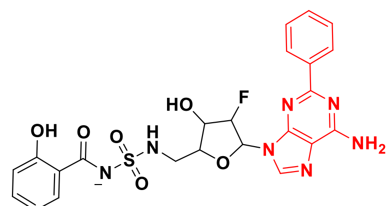
2.2.4. Base Modification. Systematic modifications of the nucleobase unit of Sal-AMS **20** and their effects on MbtA inhibition and anti-tuberculosis activity of the resulting derivatives are discussed in this section. In 2008, Neres et al.[139] reported the exploration of SARs of the purine ring. Several analogs of **20**, bearing substitutions at the N1, N3, N7, C2, C6, and C7 positions of the adenine residue, were synthesized and evaluated for their biochemical and biological properties. The study revealed that the elimination of N1, N3, or N7 in the adenine unit (**65**) led to decreased MbtA inhibitory potency and anti-tubercular activity against Mtb H37Rv in iron-deficient conditions. In contrast, removal of the amino group ($-NH_2$) at the C6 position resulted in an inactive analog (**66**), as at least one hydrogen bond donor was crucial for the activity for these types of nucleosides. The insertion of bulky substituents at the purine C6 position, including NH-cyclobutyl and NH-cyclopentyl residues, yielded compounds devoid of any MbtA inhibitory properties and anti-tubercular activity. Interestingly, compound (**67**), which included the smaller NH-cyclopropyl residue at C6, exhibited a 3.5-fold increase in MbtA inhibitory activity compared to **20** (Figure 23). Insertion of bulky aromatic substituents at the C2 position of the purine ring, e.g., 2-phenylamino, phenylethynyl, and 2-phenyl moieties, yielded compounds (**68**), (**69**), and (**70**), respectively, which were up to 24 times more potent than **20**. Molecular modeling studies corroborated these findings and showed that the active site of the MbtA carries additional space near the adenine-C2 position of the nucleosides, and large aromatic substituents, such as phenyl rings, can be accommodated in the pocket. Compounds **68**, **69**, and **70** exhibited significantly improved whole-cell anti-tubercular activity (MIC_{99} = 49 nM in iron-deficient environment) and MbtA-binding properties (K_i values ranging from 0.27−0.94 nM) compared to **20** (Figure 23).



(71) K_i (nM) = 1.5 ± 0.2
MIC (μM) = 0.098

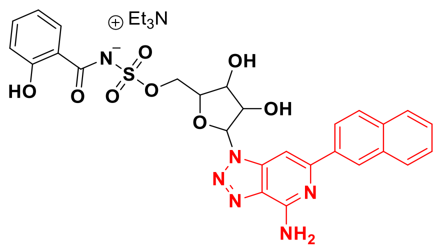


(72) K_i (nM) = 0.76 ± 0.40
MIC (μM) = 6.25

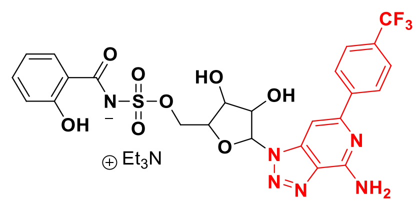


(73) K_i (nM) = 1.6 ± 0.1
MIC (μM) = 1.56

Figure 24. Chemical structures and anti-tubercular profiles of the base-modified nucleoside analogs **71**−**73**.

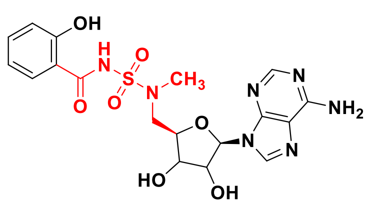


(74) K_i (nM) = 6.1 ± 0.7
MIC (μM) = 12.5

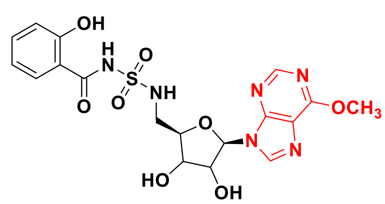


(75) K_i (nM) = 25.0 ± 1.9
MIC (μM) = >50

Figure 25. Chemical structures and anti-tubercular profiles of base-modified nucleoside analogs **74** and **75** incorporating 8-aza-3-deazaadenine as a bioisosteric replacement of the adenine ring.



(76) K_i (nM) = 158.5 ± 8.1
MIC (μM) = 0.8 ± 0.2 (-F); >64 (+Fe)



(77) K_i (nM) = 277.1 ± 12.2
MIC (μM) = 5.3 ± 1.3 (-F); >64 (+Fe)

Figure 26. Chemical structure and anti-tubercular profile of Sal-AMSNMe **76** and Sal-6-MeO-AMSN 77.

Nelson et al.[134] conducted further purine ring modifications to improve the PK profile of hit compound **21**. Among the nucleoside analogs reported by the group in 2015, three derivatives carrying base moiety alterations were explicitly designed to improve enzyme inhibition and anti-mycobacterial activities and enhance the PK profile. Substitutions at the C2 and C6-*N* positions of the adenyl group yielded N^6-cyclopropyl and N^6-cyclopropyl-2-phenyl derivatives (**71**) and (**72**), respectively, and the inclusion of an NH-cyclo-propyl moiety at C6 of the purine ring greatly enhanced the anti-tubercular activity compared to the parent compound **21**. However, nucleoside **72**, which carried a further C2-phenyl substitution, showed a 16-fold reduction in anti-tubercular activity but improved MbtA inhibitory properties (K_i = 0.76 ± 0.40 nM) compared to **21**. Additional structural modifications of the parent scaffold **21**, i.e., the inclusion of a fluorine atom at the C2′ position of the sugar unit and C8-phenyl substitution at the adenine moiety, led to analog (**73**), which exhibited improved MbtA inhibitory property with a K_i value of 1.6 ± 0.1 nM and approximately 8-fold decrease in anti-tubercular activity. These structural modifications resulted in cLogP values ranging from 0.89 (**21**) to 0.5 and 2.9 for analogs **71** and **72**, respectively (Figure 24).

Krajczyk et al.[140] further expanded the SARs of base-modified Sal-AMS analogs and designed and evaluated a series of derivatives containing the isosteric 8-aza-3-deazaadenine unit that was substituted at its C2 position with various aryl groups. The alteration of the Sal-AMS **20** adenosine scaffold led to analogs such as compounds (**74**) and (**75**), which were able to achieve MbtA inhibition with K_i values ranging from 6.1 to 25 nM and arrest the growth of Mtb with MIC ranging from 12.5 to >50 μM. These findings indicated that bioisosteric replacement of the purine ring and C2-aryl substitution led to a significant decrease in the efficacy profile of the derivatives (Figure 25).

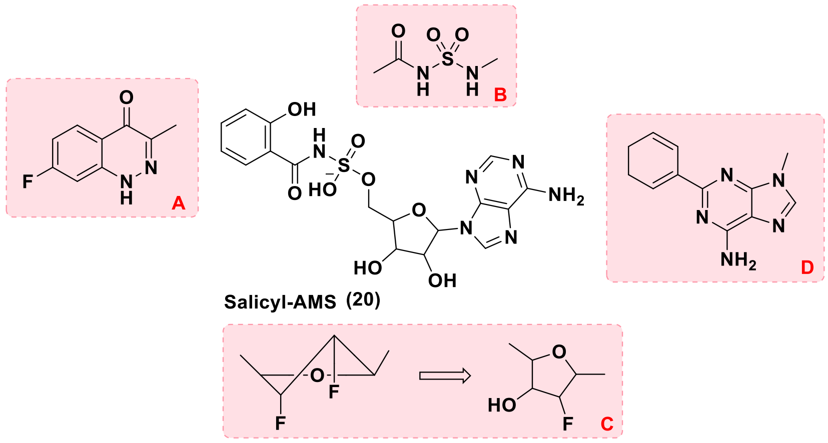


Figure 27. A conclusive summary of the structure−activity relationship on the parent scaffold of Sal-AMS **20**, a first-in-class anti-tubercular agent that targets the first step of the biosynthesis of salicyl-capped siderophores (mycobactins). The main aim behind Sal-AMS analogs development was to produce newer candidates with improved PK profile without compromising the anti-tubercular profile of Sal-AMS **20**. Based upon best structural fragmentation analysis, SARs have been elaborated in the main text.

In a joint research project between Tan and Quadri et al.,[108] two novel Sal-AMS derivatives, Sal-AMSNMe (**76**) and Sal-6-MeO-AMSN (**77**), were reported. The groups investigated the SARs of the base- and linker-modified nucleotides (Figure 26).

The compounds were tested for anti-mycobacterial activity against a strain of the non-pathogenic *Msmeg* (Δ*EM*-pMbtA$_{tb}$) engineered to be susceptible to MbtA inhibitors in a MbtA$_{tb}$-dependent manner. The study revealed that **77**, which was devoid of the purine C6 amino motif of Sal-AMS **20**, exhibited an almost 6-fold decrease in anti-mycobacterial activity compared to **20** (MIC = 0.8 ± 0.2 μg/mL in iron-depleted conditions), with MIC values of 5.3 ± 1.3 and >64 μg/mL in GAST and GAST-Fe medium, respectively. Sal-AMSNMe showed an anti-mycobacterial activity comparable to that of Sal-AMS. The study also revealed that the MbtA inhibitory efficacy of Sal-6-MeO-AMSN was significantly lower than that of **20** (K_i = 26.6 ± 2.5 nM), with a K_i value of 277.1 ± 12.2 nM, suggesting that the presence of the adenosine C6-NH_2 group in nucleoside-based MbtA inhibitors is crucial for the optimal biological activity for this class of compounds.

The major and most significant SAR findings related to Sal-AMS MbtA inhibitors are summarized in Figure 27.

(A) Replacement of the *ortho*-hydroxy aromatic group of parent compound **20** with a fluoro-substituted cinnolone unit resulted in analog **50** with potent anti-tubercular profile (K_i = 11.6 ± 2.4 nM; MIC = 2.3 μM), but most interestingly with a remarkably enhanced PK profile. Oral administration of analog **50** (25 mg/kg) in female Sprague−Dawley rats model indicated a 4 times improved half-life ($t_{1/2}$ = 40 ± 5 i.v., min), 7-fold greater steady-state volume of distribution (V_d = 0.57 ± 0.17 i.v., L.kg^{-1}), and 2 times improved oral bioavailability (F = 2.1%) compared to lead candidate **20**. Henceforth, we can conclude that replacement of the *ortho*-hydroxy aromatic group with the fluoro-substituted cinnolone moiety may encourage medicinal chemists to further develop analogs in this direction.

(B) Substitution of the sulfamate linker unit with sulfamide bridge **21** led to an increase in overall anti-tubercular profile (K_i = 3.7 ± 0.6 nM; MIC = 0.19 μM) compared to **20**. Additionally, the PK profile of **21** showed a 3-fold improved oral bioavailability (F = 3.5 ± 0.2%) and a more than 3-fold higher maximum plasma concentration (C_{max} = 0.70 ± 0.07 μg.mL^{-1}) compared to candidate **20** in a doubly cannulated female Sprague−Dawley rat model.

(C) Difluorination of the ribose unit of the nucleosides led to 2′-*endo*, 3′-*exo* conformer **56** with the most promising PK profile in cannulated rat models. The PK profile indicated a 4-fold improved oral bioavailability (F = 5.3 ± 1.3%) with 15-fold increase in maximum plasma concentration (C_{max} = 5.99 ± 0.56 μg mL^{-1}) and 25 times greater half-life ($t_{1/2}$ = 267 ± 36 i.v., min) compared to **20** in a doubly cannulated rat model. Despite having the most favorable PK profile, analog **56** showed a significant loss in anti-tubercular profile (K_i = 139 ± 10 nM; MIC = 37 μM). Analog **73** that contained a mono-fluoro-substituted glycosyl moiety exhibited a significant tubercle bacilli inhibition activity (K_i = 1.6 ± 0.1 nM; MIC = 1.56 μM) with promising PK profile showing a pronounced improvement in oral exposure through a preliminary increase in terminal elimination half-life ($t_{1/2}$ = 121 ± 18 min, i.v.) by blocking the P450-mediated metabolism.

(D) Incorporation of bulky aryl moieties, i.e., phenyl rings, at the C2 position of the adenine unit of derivative **73** resulted in enhanced MbtA inhibition properties and PK parameters.

2.3. Identification of Non-nucleoside MbtA-Binding Agents.

Targeting mycobactin biosynthesis pathway, and particularly MbtA enzymatic activity, might be a successful strategy in TB drug development. However, from a medicinal chemistry perspective, the majority of Sal-AMS derivatives designed so far, although extremely effective against Mtb in iron-depleted conditions, lack drug-like properties. These inhibitors exhibited off-target biological activity and *in vivo* toxicity, displayed poor drug metabolism and pharmacokinetic (DM/PK) profiles, and were likely to undergo the first-pass metabolism before reaching prime infection sites.

To this end, in 2019, the Brucoli group[125] took a different approach for the identification of novel MbtA-binding agents using a combination of whole-cell screening and target-based drug discovery methods. Screening of a diverse compound library, which was selected using standard drug-like parame-

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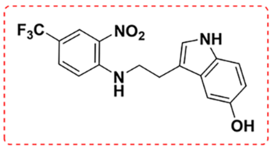
ters, i.e., MW < 500, HBD < 5, HBA < 10, rotatable bonds <10, and PSA < 120 Å, against Mtb (H37Rv), yielded a subset of H37Rv-active molecules (∼900). After selecting an appropriate MbtA ortholog, e.g., *Msmeg* MbtA (MSMEG_4516), the compounds were tested for their ability to bind to the *Msmeg* MbtA. The binding studies were performed using a robust fluorescence-based thermal shift assay, water-ligand observed gradient spectroscopy (LOGSY), and saturation transfer difference experiments to confirm active hits and eliminate false positives.

A hit compound 3-(2-((2-nitro-4-(trifluoromethyl)phenyl)-amino)ethyl)-1*H*-indol-5-ol (**78**) (Figure 28) was identified



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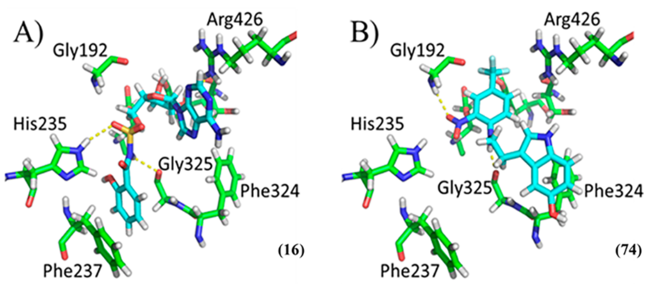

Figure 28. Structures of docked compounds Sal-AMS **20** and **78** within the *Msmeg* MbtA active site derived from PDB ID 5KEI.[103] (A) Docked conformation of Sal-AMS. (B) Docked conformation of compound **78**. Selected residues are shown as green sticks. Yellow dotted lines represent polar interactions. Structures were generated using the PyMOL Molecular Graphics System, Version 1.7, Schrödinger, LLC.

from this campaign. The compound showed anti-tubercular activity against actively replicating Mtb (MIC_{90} H37Rv = 13 μM) and the ability to bind to MbtA. Compound **78** also inhibited the growth of the Koch bacillus in Mtb-infected macrophages (intracellular MIC_{90} = 11 μM in RAW264.7), as evaluated using a live-cell fluorescence-based screen. Molecular modeling assisted docking of **78** into the proposed binding pocket of MbtA, revealing that **78** and Sal-AMS **20** might occupy different binding orientations in the enzyme.

2.4. Discovery of the C-16 AMS as MbtM Inhibitor.

Although the leading efforts to target the mycobactins biosynthetic pathway were focused on the development of MbtA and MbtI inhibitors, it is also worth emphasizing the role of the acyl-AMP ligase FadD33 (MbtM) and its potential as a target for novel *ad hoc* inhibitor design to disrupt the Mtb siderophore production.

The synthesis of the core mycobactin scaffold is performed by the enzymatic machinery encoded by the *mbt-1* gene (*mbtA-J*)[93,94] cluster. In contrast, the transfer of the lipophilic aliphatic chain to the ε-amino group of the lysine fragment (directly attached to the 2-hydroxyphenyloxazolidine moiety of the mycobactin) is performed by protein complexes encoded within the *mbt-2* gene (*mbtK-N*)[97] cluster. MbtM is responsible for the activation (adenylation) of the lipophilic acyl chain of mycobactins, which is then loaded onto the phosphopantetheinylated acyl carrier protein MbtL before the final transfer, operated by acyltransferase MbtK, to the ε-amino group on the lysine moiety (Figure 29).

A study from the Blanchard group[141] demonstrated that FadD33 has a strong affinity for long-chain saturated lipids as substrates and can be inhibited by 5′-*O*-[*N*-(palmitoyl) sulfamoyl] adenosine (**79**, C16-AMS), a structural analog of C12-adenylate (**80**) (Figure 29).[142] For this study, MbtM (FadD33) was incubated with several fatty acids of different lengths ranging from C6 to C16 units, including lauric acid, and with ATP, and C16-AMS. In the assay, significant MbtM inhibitory activity was observed by the palmitic acid fragment-containing C16-AMS nucleoside.

This might lead to the conclusion that MbtM, a bifunctional enzyme that operates similarly to MbtA, can be a promising drug target for inhibiting mycobactin biosynthesis. The only structural difference between ligands **79** and **20** is represented by the *ortho*-hydroxy aromatic ring of Sal-AMS, which is replaced by the linear carbon chain of C16 palmitic acid.

2.5. PPTase Inhibitors' Development.

In the complex process of construction of mycobactins, 4′-phosphopantetheinyl transferases (PPTase)[106] play a significant role by translationally modifying carrier proteins (CPs) with the inclusion of a 4′-phosphopantetheine (PPant) moiety. In the mycobacterial NRPS-PKS hybrid systems,[143] the mycobactin framework is constructed by the elongation of a starting monomer followed by the formation of a growing intermediate that is covalently tethered by a carrier protein onto the PPant arm through a thioester linkage.[144] Before receiving the intermediate, the carrier protein must be post-translationally converted from its inactive apo form into its active holo form. This step is catalyzed by the Mg^{2+}-dependent 4′-PPTase that transfers the PPant arm from coenzyme A onto a conserved serine residue of the CP domain through phosphodiester bonds. PPant−chemical intermediate thioester complexes are key elements involved in the secondary metabolites, e.g., mycobactins, chain elongation process in NRPS, PKS, and fatty acid synthetase machineries.[145] PPTase is an attractive anti-tubercular drug target, and its inhibition can disrupt mycobactin biosynthesis.

In search of potential PPTase inhibitors, Duckworth and Aldrich[146] first developed several potent Sfp 4′-phosphopantetheinyl transferase specific inhibitors. Compound (**81**) was the most promising ligand of the series, with a significant PPTase inhibition activity determined by both fluorescence polarization (FP) (K_i = 0.180 ± 0.01 μM) and isothermal titration calorimetry (ITC) (K_i = 0.609 ± 0.088 μM) assays. Two more ligands, sanguinarine chlorides (**82**) and (**83**), were also tested in this work. However, **82** displayed inferior activity in the FP and ITC assays. Further to this, Vickery et al.[147] solved the X-ray crystal structure of Sfp-type PPTases from two of the most prevalent mycobacterial species, *M. tuberculosis* and *M. ulcerans*, and identified specific PPTase inhibitors in their work. Among the small library of inhibitors evaluated in the study, ligand **82** was the most potent inhibitor against the target proteins, with IC_{50} values ranging from 4.9 ± 0.2 to 22 ± 2 μM. Rohilla et al.[148] carried out a docking and cheminformatics study to identify some potent PPTase inhibitors using Autodock 4.2 and Glide platforms. The ligand PS-40 (NSC-328398, **83**) emerged as the most potent inhibitor, with IC_{50} and MIC_{90} values of 0.25 and 10 μg/mL, respectively. Negligible cytotoxicity was observed against three mammalian cell lines with the selectivity index greater

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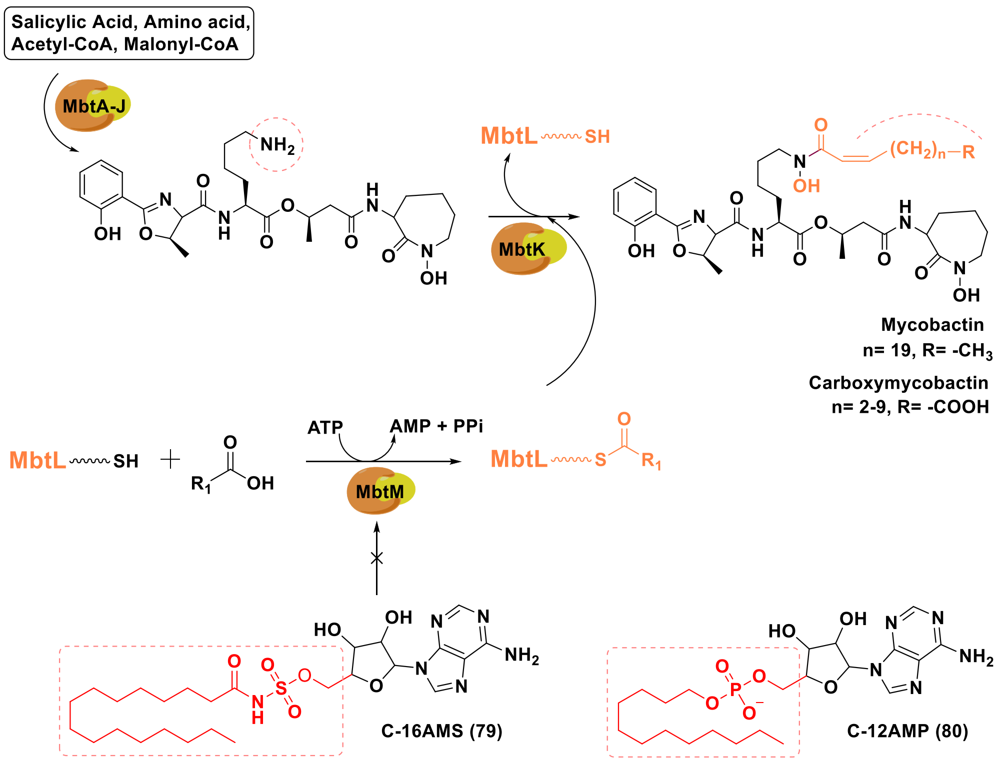


Figure 29. Graphical representation of the significant role of MbtM in the mycobactin biosynthesis pathway. C16-AMS **79** was found to inhibit the MbtM enzyme.

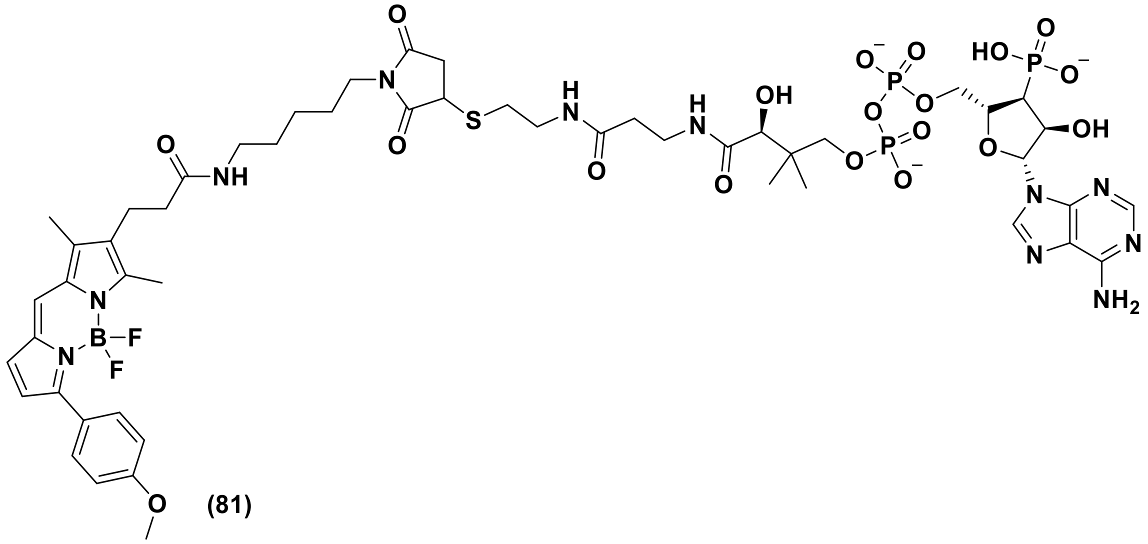
than 10. Grunewald et al.[149] prepared a unique antibody−drug conjugate system (ADC, **84**) utilizing 11- and 12-mer amino acidic substrates of phosphopantetheinyl transferase as warheads to be linked to the antibody. Using Sfp-PPTase, 63 sites could be efficiently labeled with an auristatin toxin, resulting in 95 homogeneous ADCs. 11- and 12-mer PPTase substrate sequences were inserted at the 110 constant region loop positions of trastuzumab IgG. Insertion of 11- and 12-mer peptide tags impacted the thermal stability and PK profiles. The peptide tag between residues 118 and 215 was labeled as the CH1 domain, where residues 228−340 and residues 341−447 spacing indicated the CH2 and CH3 domains, respectively. The melting temperature (T_m) values of antibodies with insertions in the CH1 and CH3 domains were on average 1.4−4.2 °C lower than that of trastuzumab (T_m = 69.7 °C). In the case of the CH2 domain, the average value of the construct was 13.3 °C, indicating that the insertion site strongly modulated thermal stability irrespective of the drug molecule. The ADCs labeled in the CH2 domain displayed an excellent PK profile due to the faster clearance from the circulation with negligible drug loss. This approach resulted in a site-specific drug delivery system with considerably better PK profile and an interesting *in vitro*/*in vivo* profile (Figure 30).

Nucleoside analogs are designed to mimic endogenous substrates and inhibit the activity of target enzymes involved in the mycobactin biosynthesis. A series of non-specific mycobactin biosynthesis inhibitors and non-conventional anti-tubercular drug delivery systems proposed to disrupt mycobacterial siderophore synthesis are reported below.

2.6. Non-specific Mycobactin Biosynthesis Inhibitors.

Several analogs structurally related to the mycobactins were designed, synthesized, and evaluated against Mtb under iron-deprived (GAST) and iron-rich conditions (GAST-Fe) for a comparative assessment. The primary assumption behind this synthetic strategy was that mycobactin-mimicking compounds with structural features of mycobacterial siderophores might disrupt the ferri-mycobactin transport systems across membranes, inhibit enzymes related to siderophores biosynthesis, and arrest bacterial growth under iron-deprived conditions. Three classes of mycobactin analogs have been designed so far based on this hypothesis. Their molecular characteristics and anti-tubercular activity are described in the next section.

2.6.1. Pyrazoline Analogs. In 2008, Stirrett et al.[150] synthesized a library of small molecules with structural similarities to the mycobactins framework and evaluated them against Mtb. Their basic structural scaffold comprised the 3,5-diaryl-1-carbothioamide-pyrazoline motif, which resembled the hydroxyphenyl-oxazoline unit of the mycobactin and carboxymycobactin siderophores. The compounds were not target specific and were evaluated for their ability to arrest the growth of Mtb in both iron-deficient and iron-rich media and to inhibit a salicylation enzyme targeted by Sal-AMS **20**. The bactericidal compound (**85**) displayed a very high anti-tubercular activity in the series, with IC_{50} and MIC_{90} values of 8 and 21 μM, respectively, in iron-deprived conditions. Analog **85** showed no cytotoxicity against HeLa cells (CD_{50} = 398 μM), and the candidate was not active against Mtb (MIC_{90} = 333 μM) in iron-rich media GASTD+Fe, thus indicating its involvement in mycobactins system functionalities, with a remarkable selectivity index (SI_{Mtb} = $IC_{50GAST\text{-}D}/IC_{50GAST}$) value of 15. The scaffold of analog **85** provided a promising platform for lead design against CETs and for exploring the mycobactin synthetic pathway. In continuation of this work, in 2011, Ferreras et al.[151] designed two libraries of mycobactin analogs containing diaryl-substituted pyrazoline (DAP) and (2*E*)-2-benzylidene-*N*-hydroxyhydrazine carboxamide



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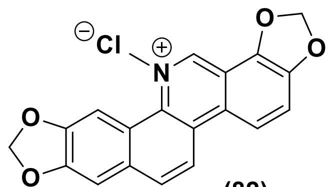

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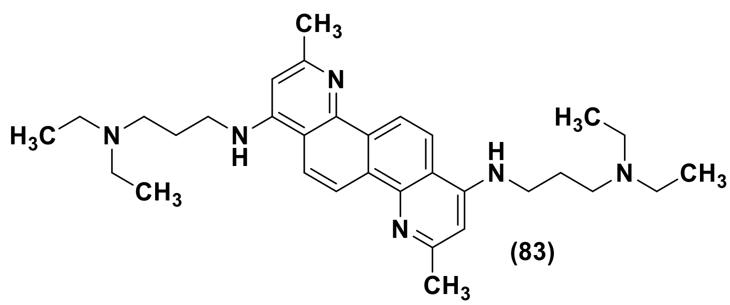

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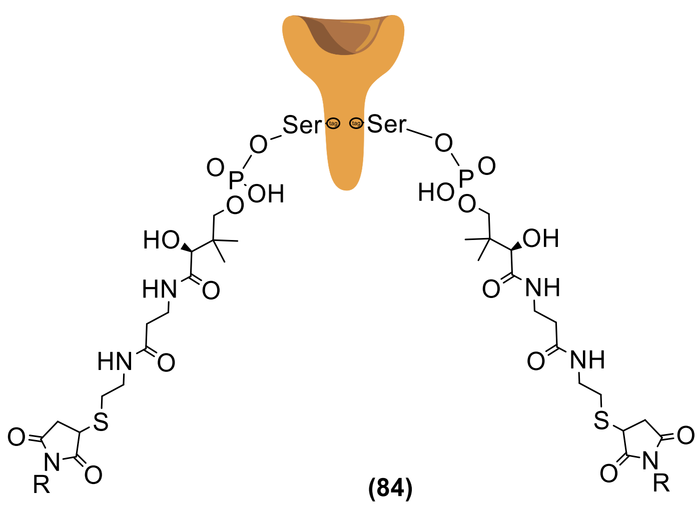

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Figure 30. Chemical structures of the PPTase inhibitors **81−83** and representation of antibody−drug conjugate system (ADC) **84** utilizing amino acidic substrates of phosphopantetheinyl transferase as a payload warhead.

(BHHC)-based scaffolds. The DAP and BHHC derivatives were evaluated against Mtb and *Yersinia pestis*. Among the active DAP derivatives, compounds (**86**) and (**87**) were the most potent against Mtb (IC_{50} = 4−7 μM, MIC = 16 μM). Among the BHHC derivatives series with defined IC_{50} and MIC values, (**88**) and (**89**) were the most active against Mtb (IC_{50} = 2−4 μM, MIC = 6−10 μM). These two compounds

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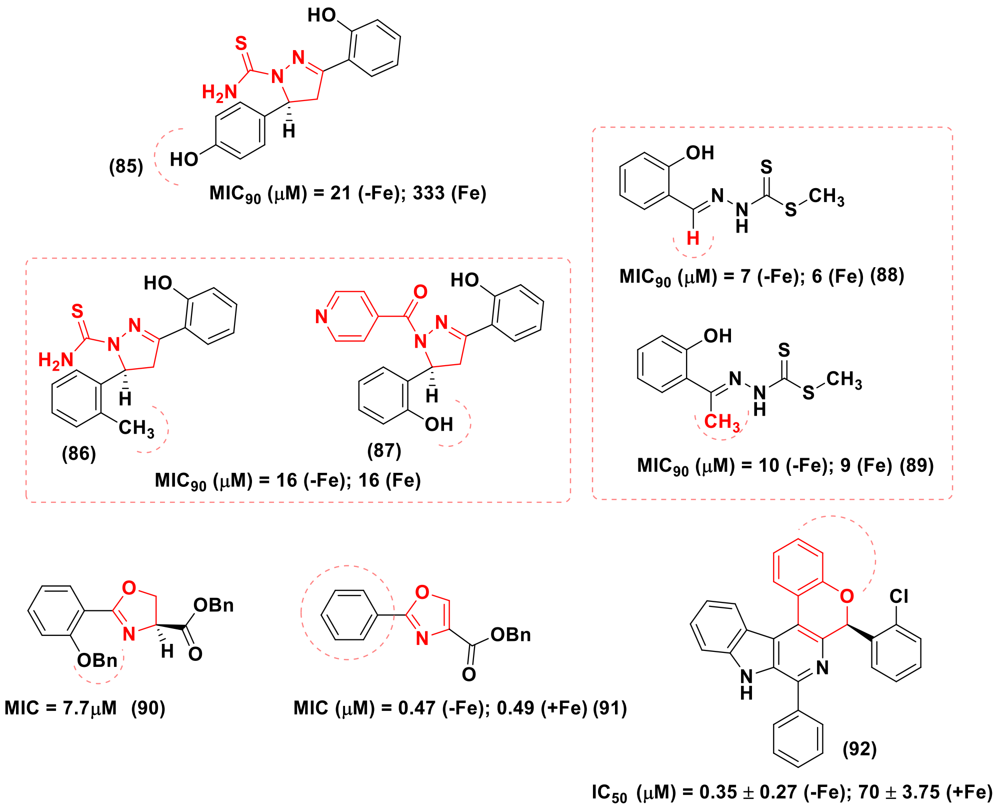

Figure 31. Non-specific mycobactin inhibitors **85**−**92** showing significant anti-tubercular activity against Mtb cultured in iron-rich media. Compounds **86**−**89**, whose structures included mycobactin-resembling units, exhibited anti-TB activity irrespective of the iron content environment. These agents were not designed with a specific enzyme as the target. However, derivatives **90** and **91** that showed remarkable anti-TB activity in iron-deficient conditions might impair siderophore-related transport systems and enzymatic processes by virtue of their molecular frameworks, which incorporated structural features resembling portions of the mycobacterial siderophores scaffolds, i.e., the salicyl-capped oxazoline component.

displayed no significant cytotoxicity in mammalian cells at their respective MIC values.

However, the majority of active compounds exhibited similar anti-tubercular activity in both iron-depleted (GAST) and iron-replete (GAST+Fe) media, suggesting that the compounds' targets might be linked to mycobacterial processes essential in both low and high iron conditions (Figure 31).

2.6.2. Oxazole and Oxazoline Analogs. Moraski et al.[152] in 2010 also designed a >100-member library of mycobactin analogs utilizing the oxazoline and oxazole scaffolds to mimic the hydroxyphenyl-oxazoline motif of the Mtb siderophores. The study rationale was based on previous findings related to ND-005859[153] (**90**), a non-iron-binding, dibenzyl-protected oxazoline molecule, which is a precursor in the synthesis of mycobactin S and T (Figure 5). Compound **90** exhibited a remarkable anti-tubercular activity (MIC = 7.7 μM) and was non-cytotoxic against Vero cells (IC$_{50}$ > 128 μM). Interestingly, after deprotection of the benzyl groups, the resulting ND-005862 analog, which possessed iron-chelating properties, lost the anti-tubercular activity of the parent compound completely. From this observation, the group explored the SAR around hit compound **90** and synthesized easily accessible derivatives bearing the benzyl ester unit that were tested *in vitro* against Mtb in iron-deprived conditions and *in vivo* to assess toxicity using BALB/c mice.

A large number of these oxazole derivatives showed significant anti-tubercular activity, in particular compound (**91**), which had a MIC value of <0.5 μM against replicating H37Rv and was not cytotoxic in Vero cells (IC$_{50}$ = 121 μM). However, the metabolic instability of the benzyl esters prevented their use as orally active pre-clinical candidates (Figure 31).

2.6.3. Chromane Derivatives. In 2012, Veerepalli et al.[154] designed and synthesized chromeno[4,3-*d*]benzimidazo [1,2-*a*] pyrimidine analogs as siderophore inhibitors. The compounds contained a 2-phenyl-3,4-dihydro-2*H*-chromene (flavan) skeleton that mimicked the 2-hydroxyphenyl-oxazoline moiety of mycobactins. Among the designed derivatives, analogs that contained chloro substitutions on the aromatic ring directly attached to the benzopyran scaffold displayed the highest potency with better inhibitory selectivity toward Mtb grown in an iron-depleted medium. The benzopyran moiety having a chloro-phenyl substituent (**92**) showed significant activity against Mtb (H37Rv) in iron-deficient medium with IC$_{50}$ values of 0.35 ± 0.27 μM, whereas, in iron-rich conditions, the derivatives showed poor anti-mycobacterial activity (IC$_{50}$ = 70 ± 3.75 μM) (Figure 31).

Canonical medicinal chemistry strategies were adopted to target the Mtb siderophores systems with Sal-AMS nucleoside analogs and mycobactin-mimicking molecules. However, an alternative approach, which involved the synthesis of a heterodimeric mycobactin-conjugated drug delivery system, has also been adopted to disrupt iron metabolism in mycobacteria. This unique drug delivery system was successfully employed to impair iron chelation and uptake mechanisms in drug-sensitive and MDR Mtb strains.

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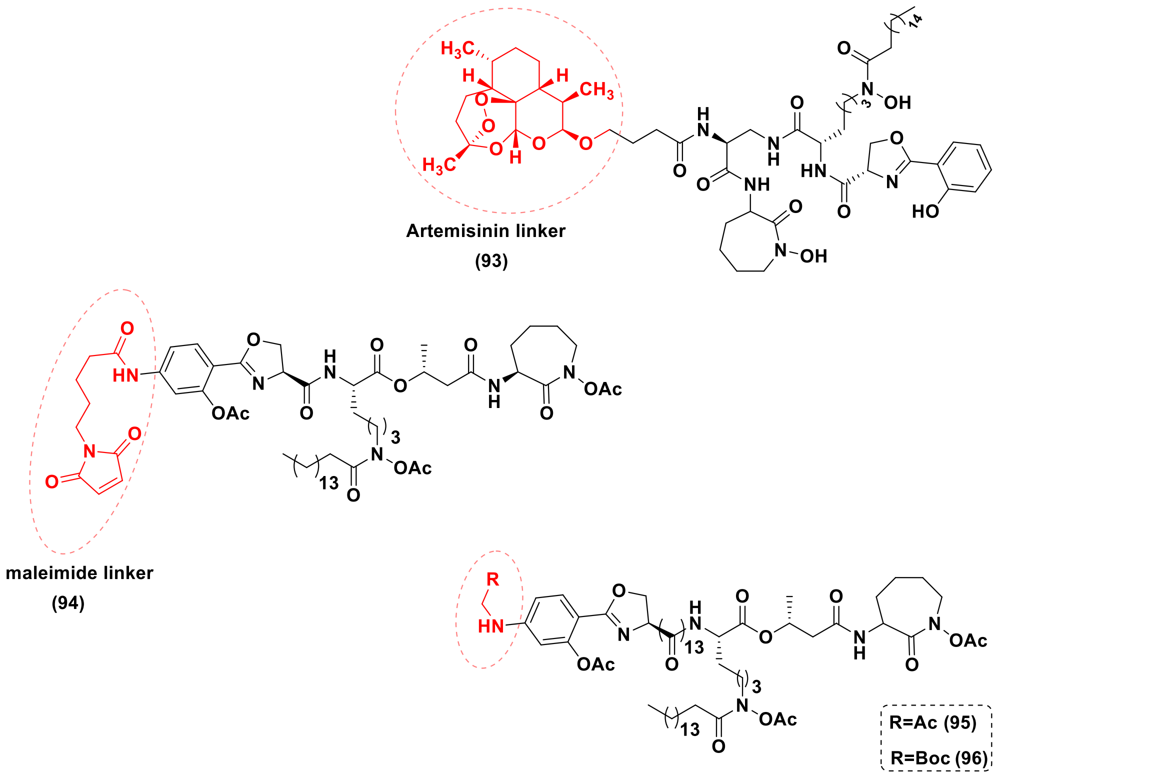

Figure 32. Hybrid siderophore−drug conjugates developed by Miller et al. The mycobactin−artemisinin (in red) heterodimer **93** exhibited remarkable anti-malarial and anti-tuberculosis activity. Siderophore−maleimide conjugates **94−96**, which were synthesized through a thiol−maleimide reactive system, displayed improved anti-tubercular activity compared to **93**.

2.7. Alternative Heterodimeric Anti-tubercular Drug Delivery Systems: The Trojan Horse Approach. The heterodimeric hybridization represents an advanced and inventive molecular technique that involves the covalent linkage of two antibiotic pharmacophores, with a known mechanism of action, to form one molecule that might inhibit different targets within a bacterial cell.[155] In addition to this, the molecular heterodimerization approach can be coupled with the "Trojan horse" strategy to introduce the antibiotic of interest into the pathogen cell via siderophore receptors on the bacterial outer membrane. In this instance, the siderophores, i.e., mycobactins covalently linked to an anti-microbial agent, are used as vectors to transport antibiotics within the cell. This is a well-documented approach and has even been developed by various bacteria to kill other microorganisms.[156]

In 2011, Miller and co-workers[157] used the Trojan horse approach to design the mycobactin−artemisinin heterodimer conjugate (**93**), which exhibited remarkable activity against both Mtb (H37Rv) and the malaria plasmodium (Figure 32). The iron-chelating component (i.e., mycobactin) of hybrid molecule **93** activates the antibiotic activity of the artemisinin moiety, which requires iron to reduce, and then cleaves its peroxide bond. Therefore, the reduction of ferric (Fe^{3+}) to ferrous (Fe^{2+}) iron carried within the hybrid scaffold **93** might lead to Fenton radical reactions with the formation of reactive oxygen species that negatively impact the survival of the mycobacteria. The mycobactin−artemisinin conjugate **93** improved the anti-bacterial and anti-parasitic activity of artemisinin, which is not active against Mtb as a single agent and exhibited significant anti-tubercular activity against H37Rv (MIC = 0.39 μg/mL), MDR (MIC = 0.16−1.25 μg/mL) and XDR strains (MIC = 0.078−0.625 μg/mL) of Mtb.

In 2012, further studies by Miller's group[158] reported the synthesis of three mycobactin T analogs (**94−96**) bearing a mixture of *tert*-butyloxycarbonyl (Boc) and acetate (Ac) protecting groups. Compound **94** contained an additional maleimide unit, which was linked through a valerate residue to a newly installed C4 amino group in the aromatic ring (salicyl-capped unit) of the mycobactin core. Derivatives **94−96** were more potent than **93** and showed excellent anti-tubercular activity, with MIC_{90} values ranging from 0.02 to 0.88 μM against Mtb H37Rv strains cultured in 7H12 and GAS media. The compounds were selective toward Mtb, with no inhibitory effects against a panel of Gram-positive and Gram-negative (Figure 32).

3. CONCLUSION

Despite significant progress in clinical drug candidate development for TB treatment in the past 10−15 years, TB continues to be a major health burden in developing countries. The hunt for drug candidates that inhibit novel targets remains to be a pressing area of research. Studies toward a deeper understanding of Mtb biology have resulted in some progress, uncovering new therapeutic targets. Notably, in this Perspective, we have shed light on the mycobactin biosynthesis complex responsible for the essential iron transportation in Mtb. Interference with this conditionally essential pathway could be an attractive strategy for discovering anti-tubercular therapy. Iron is an ideal candidate for a redox catalyst in several cellular processes due to its ability to switch between two oxidation states. It is an essential nutrient for most microbes.[159,160] It has been observed that the impairment of mycobactin biosynthesis and iron acquisition process directly affects mycobacterial virulence and its survival in the host.[107,86]

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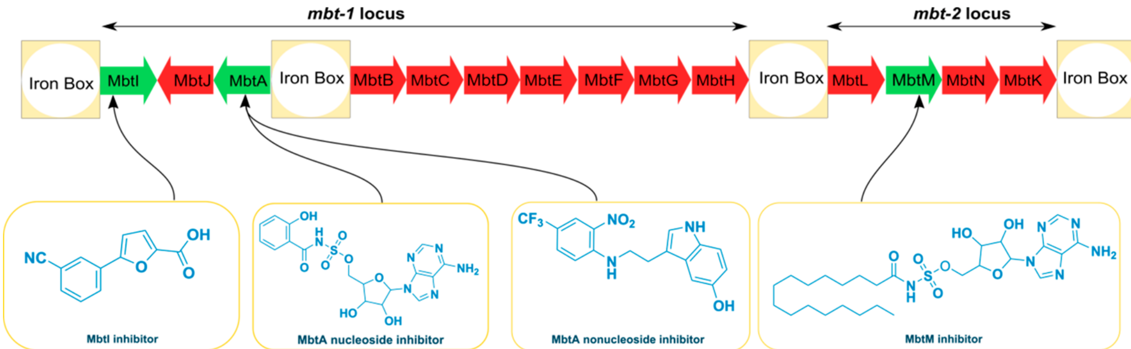

Figure 33. *mbt* gene cluster provides a golden opportunity to accelerate enzyme-based anti-tubercular drug discovery. Enzymatic targets MbtA, MbtI, and MbtM already have been shown to be promising anti-tubercular drug targets for novel TB drug design and development.

In addition, the absence of this pathway in human hosts minimizes the risk posed by off-target effects. Any resistance mechanisms in these targets are not known yet.

Structure-based rational design of MbtI and MbtA inhibitors has thus far shown promising results. In regard to salicylate synthase (MbtI) inhibitors, the development of TS inhibitors was the first stepping stone. Further development in this direction did not yield any significant candidate except C2, C3-hydroxybenzoate analog **7**, with a K_i value of 11 μM. High-throughput virtual screening led to the identification of small-molecule heterocyclic inhibitor **12** with an IC_{50} value of 5.89 μM. Thereafter, research in furan analogs development produced candidate **18** (K_i = 3.1 ± 1.0 μM, IC_{50} = 6.3 ± 0.9 μM) as the most promising MbtI inhibitor to date. For Sal-AMP ligase (MbtA) inhibitors, bioisosteric replacement of phosphate in the linker region of Sal-AMP (**19**) resulted in a potent Sal-AMP ligase inhibitor, Sal-AMS (**20**). Further modifications were attempted in the linker region, salicylic aromatic ring, sugar moiety, and nucleobase unit to improve the weak PK profile of nucleoside **20**. The replacement of the *ortho*-hydroxy functional group on the salicylic ring with 7-fluorocinnolone resulted in analog **50**, and insertion of a fluorinated ribose moiety and a C-2 phenyl substituent in the adenine unit led to derivative **73** that showed an improved PK profile compared to the parent compound **20**. Moreover, the discovery of ligand **78** prompted further research toward small inhibitors development as potential MbtA inhibitors. To date, C16-AMS (**79**) is the only candidate identified as an MbtM inhibitor. In the case of potent MbtI inhibitor **18** and MbtM inhibitor **79**, a PK profile has not yet been generated, and further studies, including PK profile development, are still required for non-specific mycobactin biosynthesis inhibitors.

Inhibitors for other enzymes in the mycobactin biosynthetic pathway are yet to be explored. Structure-based inhibitor design is a feasible approach. However, the real challenge is to establish a *proof-of-concept* that requires the respective enzymes to be recombinantly expressed. It is also necessary to establish *proof-of-concept* in an infected animal model. The design of an enzyme inhibition assay with the multimodular enzymes is another critical bottleneck. Success in continuous tailoring of multimodular NRPS proteins to synthesize heterocyclic ring systems and newer antibiotics provides hope that these techniques may offer enzyme inhibition study platforms for unexplored enzymes in the mycobactin biosynthetic pathway.[161,162]

Small molecules containing heterocycles, such as pyrazoles, oxazoles, and coumarins, mimicking the structure of 2-hydroxyphenyl-oxazoline have been designed previously. These compounds inhibited the growth of Mtb under iron-deprived conditions. However, no attempt was made to date to identify the target proteins. Further research in this direction is recommended to expedite anti-tubercular drug discovery to cope with the rising concern related to resistant strains. In continuation with our previous efforts,[125,150,151] we are currently in the process of developing target-based inhibitors with favorable biological and PK profiles.

The significant role of the conditionally essential 14 enzymes of the *mbt* gene cluster may encourage medicinal chemists to develop new lead compounds with a novel mode of action to counter the drug-resistant strains of the tubercle bacilli (Figure 33). At the same time, the biological and biochemical vulnerabilities of each enzymatic target produced by the *mbt* gene cluster must be considered for the identification of the most chemically tractable NRPS-PKS protein targets of Mbt. Understanding endogenous interactions between prospective drugs with their therapeutic targets might pave the way for the development and implementation of novel and innovative approaches to antibiotic discovery.

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The authors declare no competing financial interest.

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Barij Nayan Sinha received his M.Pharm. from Banaras Hindu University, India, and Ph.D. from Birla Institute of Technology, Mesra, India. He is currently working as a Professor in the Department of Pharmaceutical Sciences and Technology, BIT Mesra. He has been the principal or co-investigator on 10+ funded projects, including three international collaborations, two (Indo-German (DBT-BMBF) Program) on topics of anti-viral agents for the treatment of Chikungunya and Dengue viral infections and another (UGC-UKIERI) in the area of anti-cancer agents. His research interests include anti-viral and anti-cancer drug discovery and natural product chemistry using modern analytical techniques.

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■ ABBREVIATIONS USED

atpE, ATPase epsilon; CET, conditionally essential target; CPs, carrier proteins; CUEs, chorismate-utilizing enzymes; DAc, deacetyltransferase; DAP, diaryl-substituted pyrazoline; DHFR, dihydrofolate reductase; DM/PK, drug metabolism/

pharmacokinetics; DosR, dormancy survival regulon; FP, fluorescence polarization; GAST medium, glycerol-alanine-salts medium; ITC, isothermal titration calorimetry; KatG, catalase-peroxidase; MDR, multi-drug-resistant; Mtb, *Mycobacterium tuberculosis*; NRPS, non-ribosomal peptides synthetases; Pat, protein lysine acetyltransferase; PDB, Protein Data Bank; PDIM, phthiocerol dimycoceresates; PKS, polyketide synthase; pncA, pyrazinamidase/nicotinamidase; PPTase, phosphopantetheinyl transferase; Rec, receptor protein; RNIs, reactive nitrogen intermediates; Rpf, resuscitation promoting factor; rpsL, ribosomal protein S12; rrs, ribosomal RNA; Sal-AMP, salicyl-adenosine monophosphate; Sal-AMS, 5′-*O*-*N*-salicylsulfamoyl adenosine; TDR, total drug-resistant; *thyA*, thymidylate synthase; TS, transition state; WHO, World Health Organization; XDR, extensively drug-resistant

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